

Calling Bullsh*t on Data Visualisations!

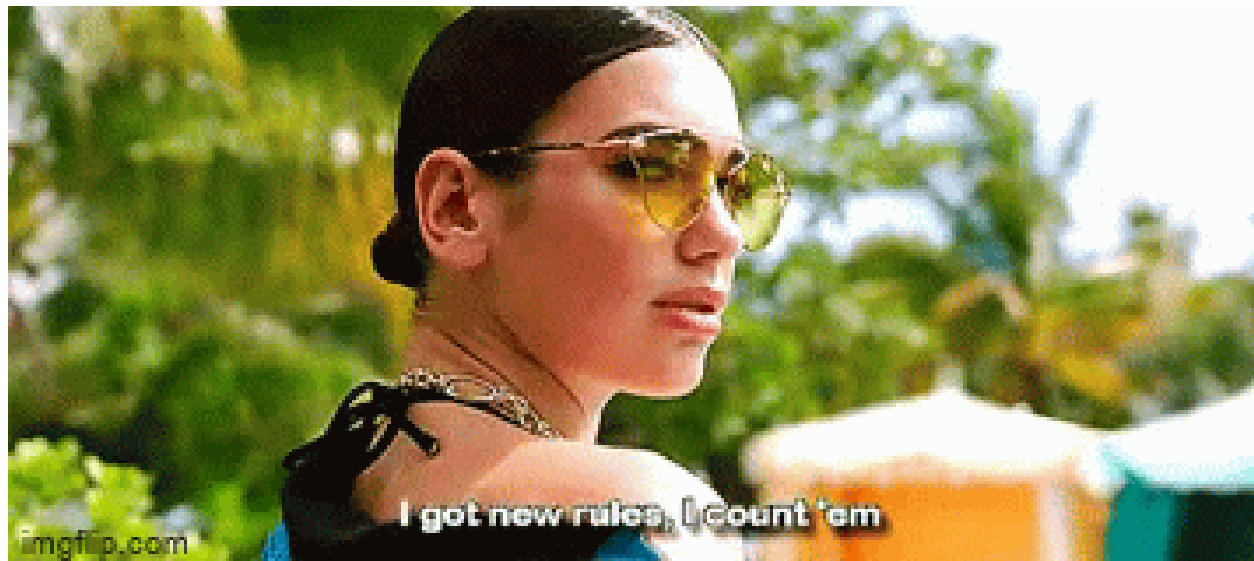
CM4125 – Topic 3

But before... let's see what you got last week!

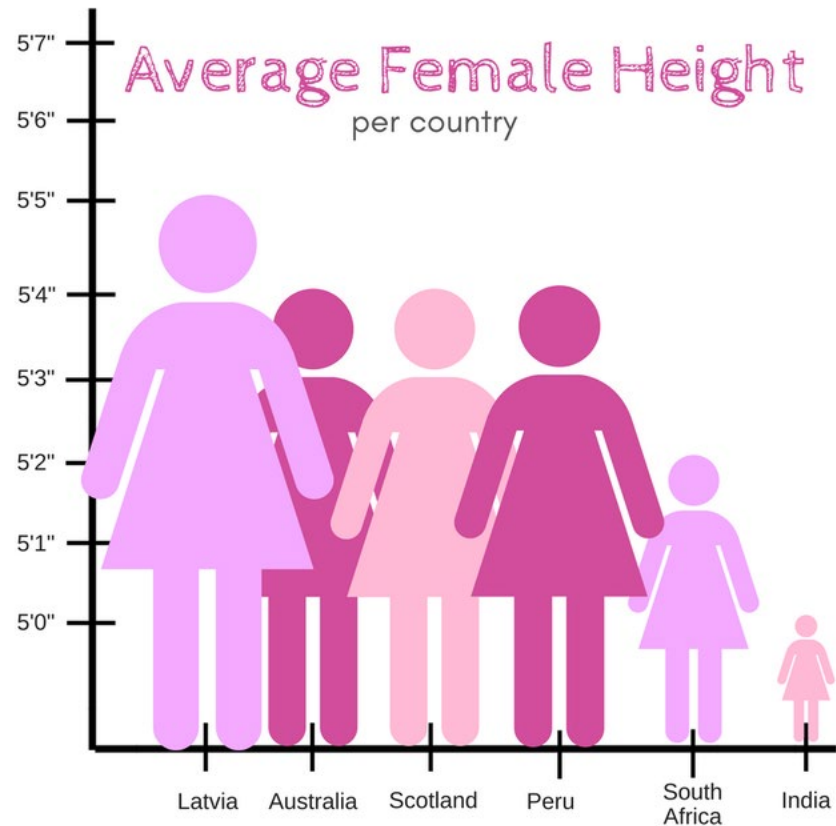
What is this lecture about?

- Based on Week 6 of the Calling Bullsh*t course of the University of Washington by Carl T. Bergstrom and Jevin West
 - <https://www.callingbullshit.org/syllabus.html#Visual>
- Published as a book!
 - <https://www.amazon.co.uk/Calling-Bullshit-Skepticism-Data-Driven-World/dp/0525509186>
- Supplementary reading:
 - Edward Tufte (1983) *The Visual Display of Quantitative Information*. Chapters 2 (Graphical integrity) and 5 (Chartjunk: vibrations, grids, and ducks).
 - Alberto Cairo (2019) *How Charts Lie: Getting Smarter about Visual Information*. W.W. Norton and Company.

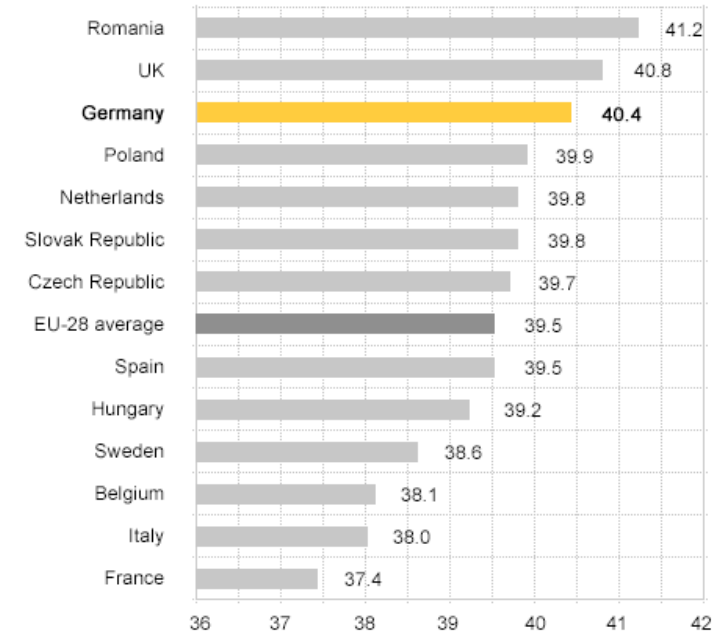
The rules



1. Bar chart y-axes (dependent variable) need to include zero

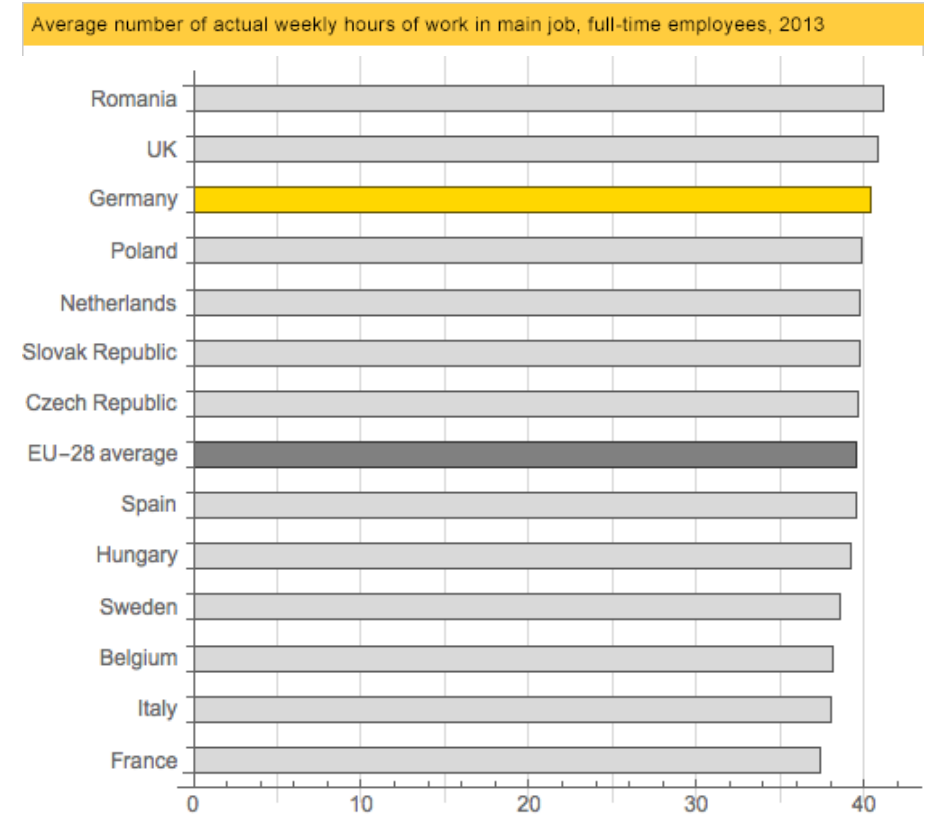
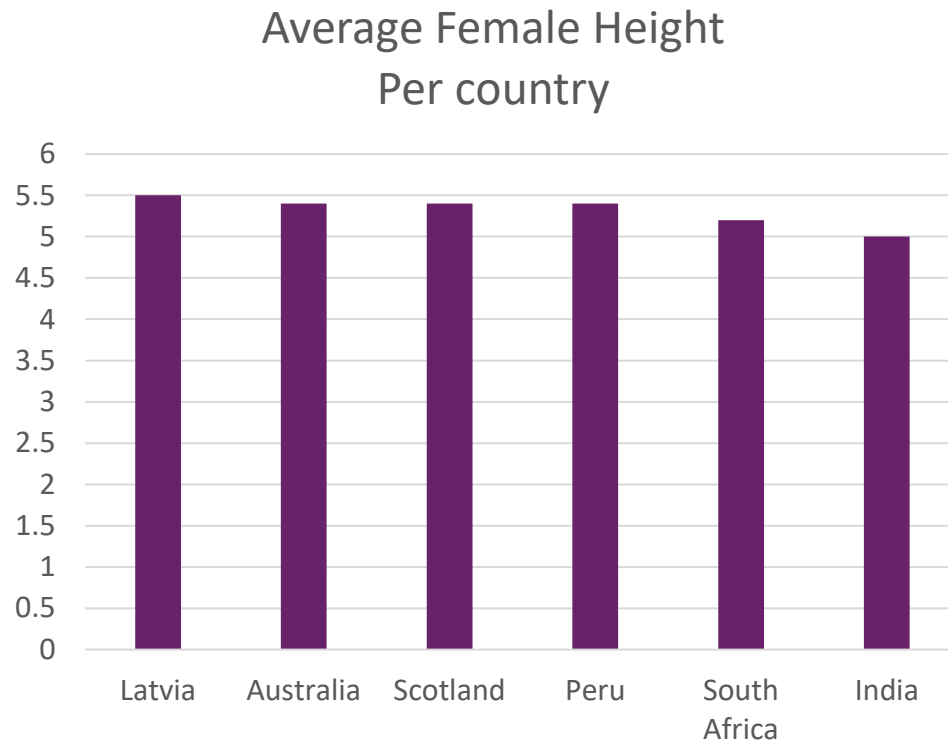


Average number of actual weekly hours of work in main job, full-time employees, 2013



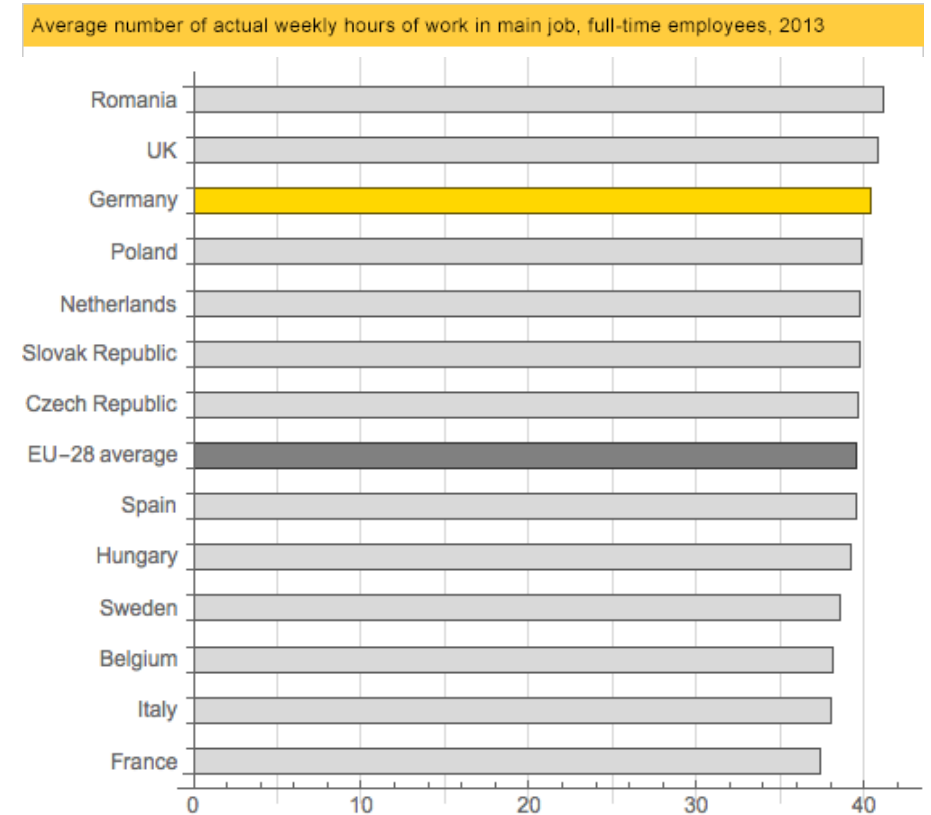
Source: Eurofound 2014

1. Bar chart y-axes (dependent variable) need to include zero



1. Bar chart y-axes (dependent variable) need to include zero

- I know, it reduces the “impact” of the plot, but this bar chart is more trustworthy!
- Moreover, sometimes verbal number stories are “better” (i.e. more appropriate) than bars!
 - In this case, maybe you just want to say that Germany is 0.9 over the EU average, and there’s no need for pictorial stories!
- OR show this info in a table!

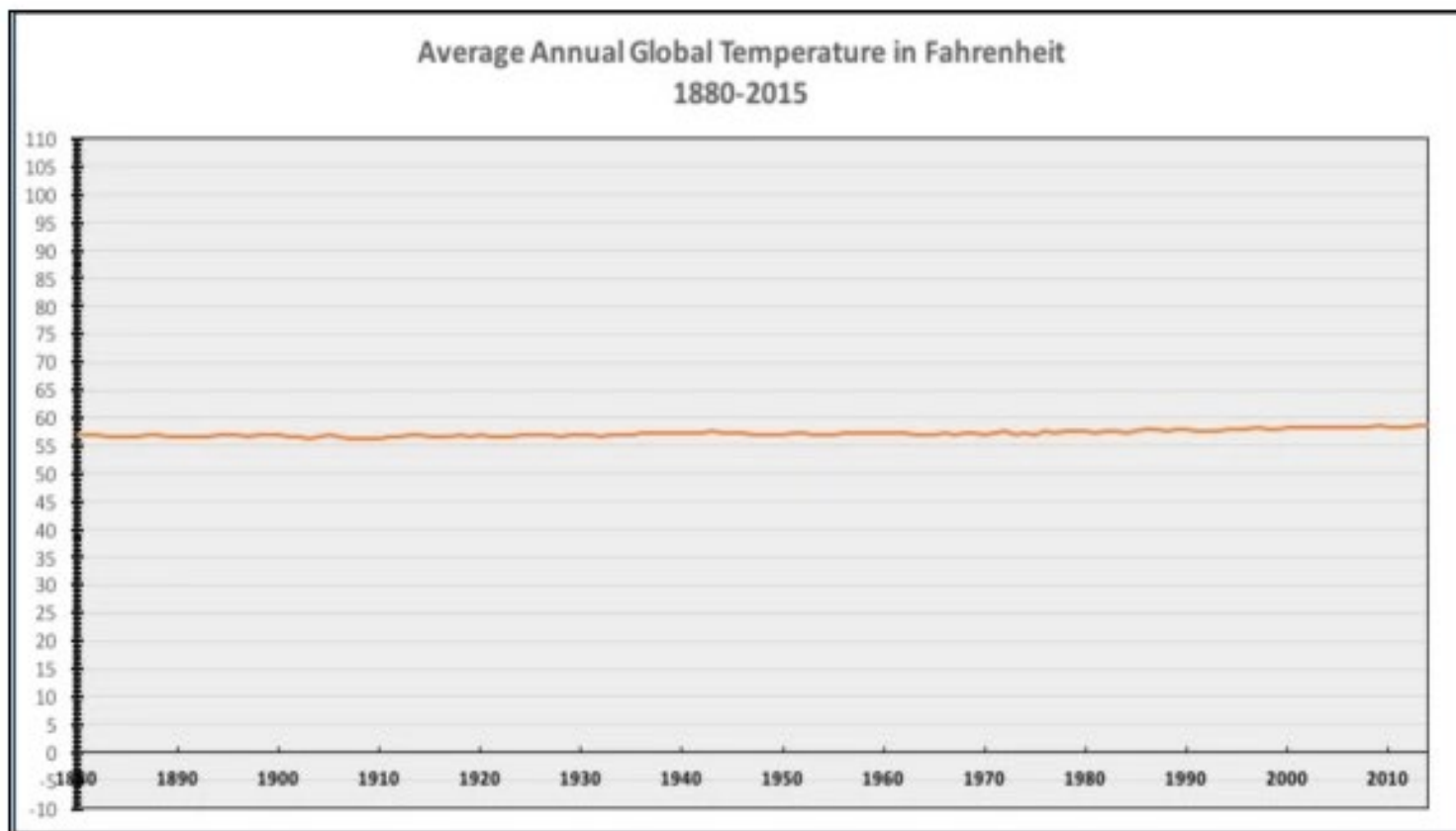


- In cases like this, tables are much better!

Table 1. Average number of actual weekly hours of work in main job, full-time European employees, 2013. Highest values marked in **bold**.

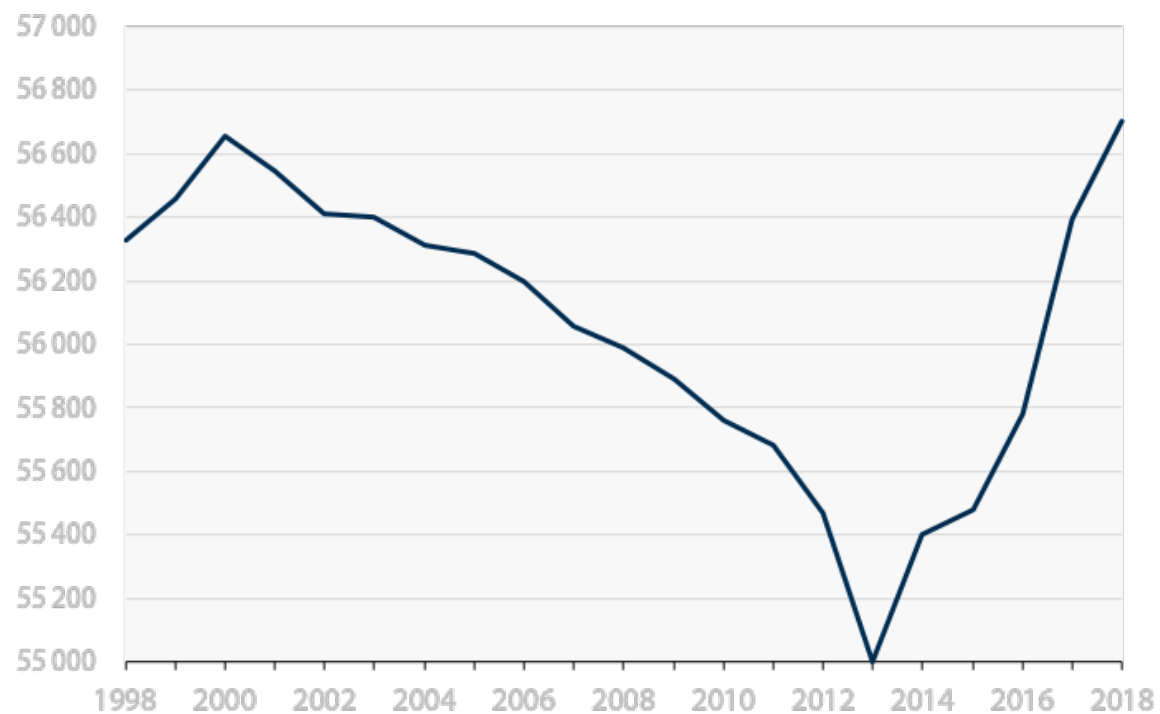
Country	Population 2013 (millions) ↑	Avg. Weekly Hours ↑
Romania	19.98	41.2
UK	64.14	40.8
Germany	80.65	40.4
Poland	38.04	39.9
Netherlands	16.8	39.8
Slovak Republic	5.41	39.8
Czech Republic	10.51	39.7
EU-28 Average	505.7	39.5
Spain	46.6	39.5
Hungary	9.87	39.2
Sweden	9.6	38.6
Belgium	11.16	38.1
Italy	60.31	38.0
France	66	37.4

2. Line graph y-axes don't need to include zero

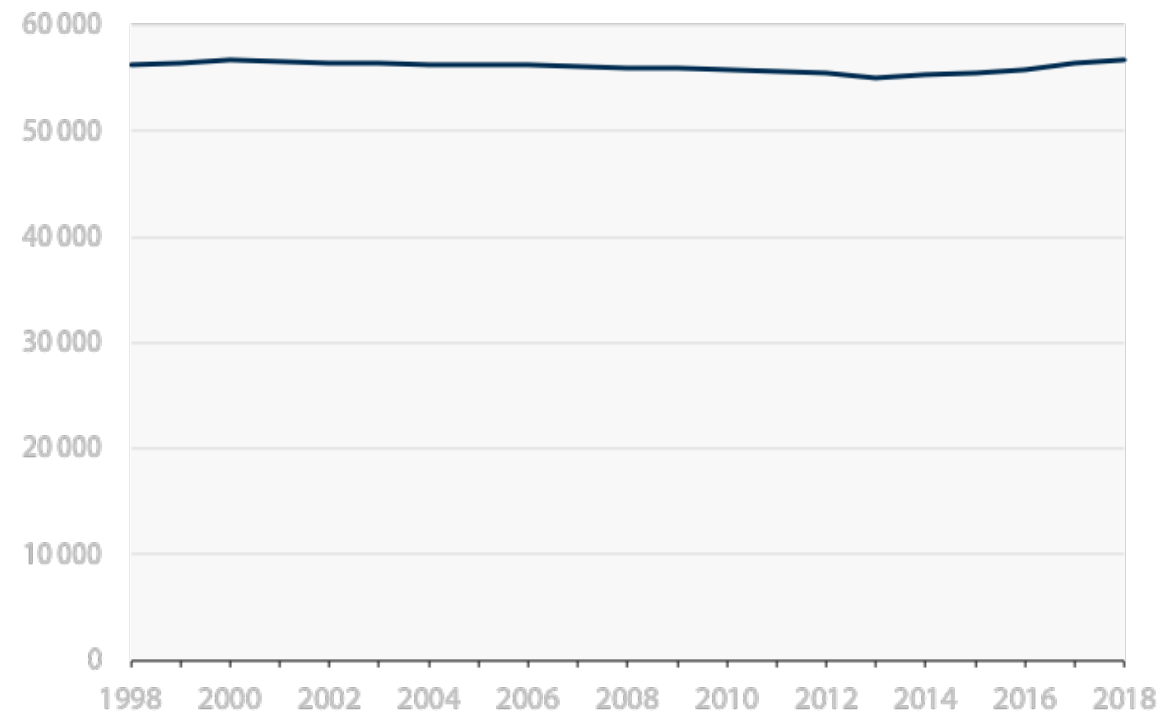


Well...

Sales of widgets

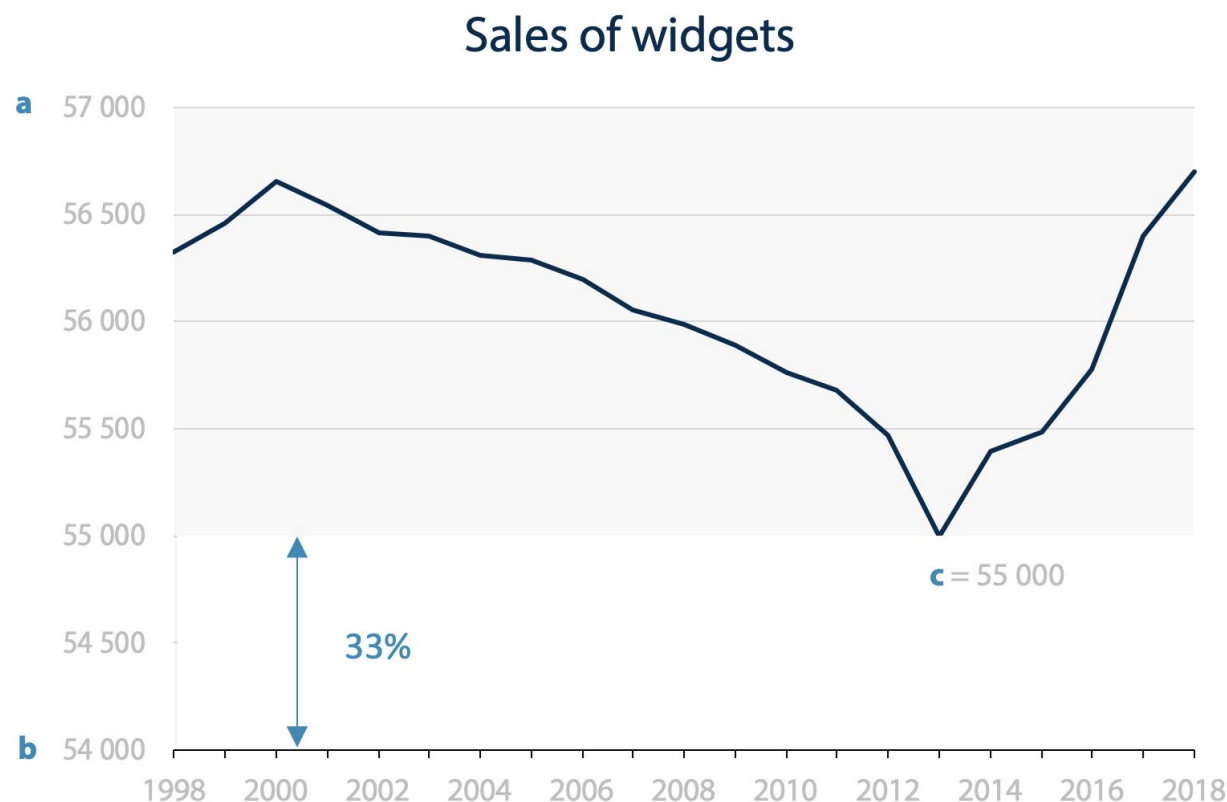


Sales of widgets



Solution

Empty space = $\frac{\text{Minimum data value} - \text{Minimum scale value}}{\text{Maximum scale value} - \text{Minimum scale value}}$



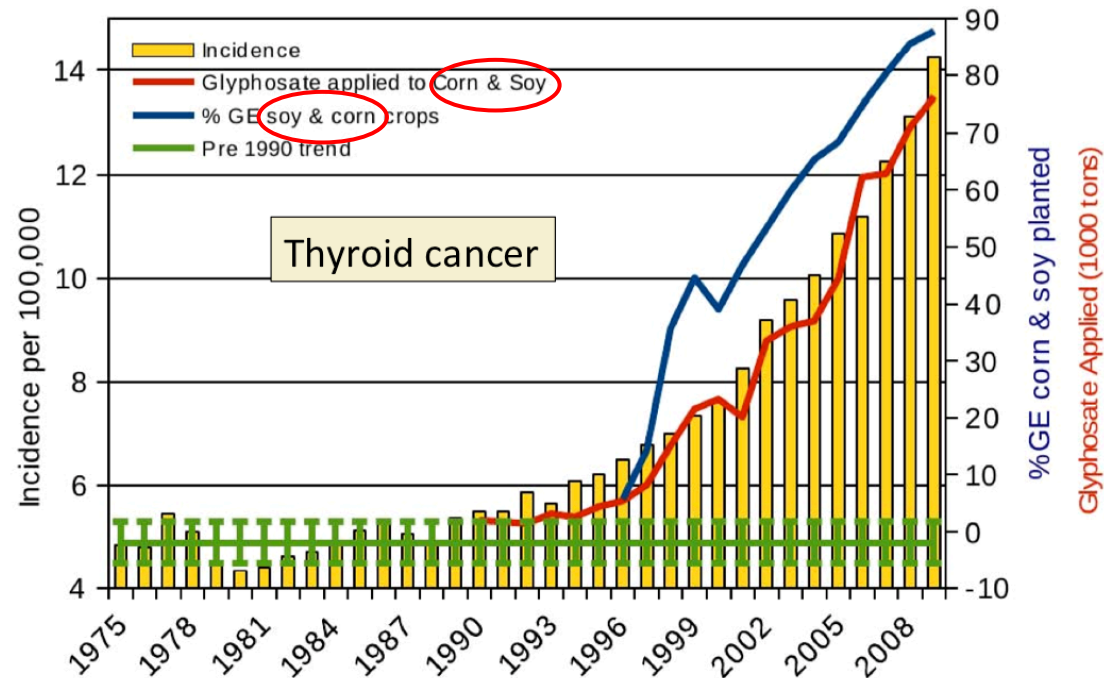
$$x = \frac{(c - b)}{(a - b)}$$

$$b = \frac{3c - a}{2}$$

$$b = \frac{3 * 55,000 - 57,000}{2} = 54,000$$

3. No multiple axes on a single graph

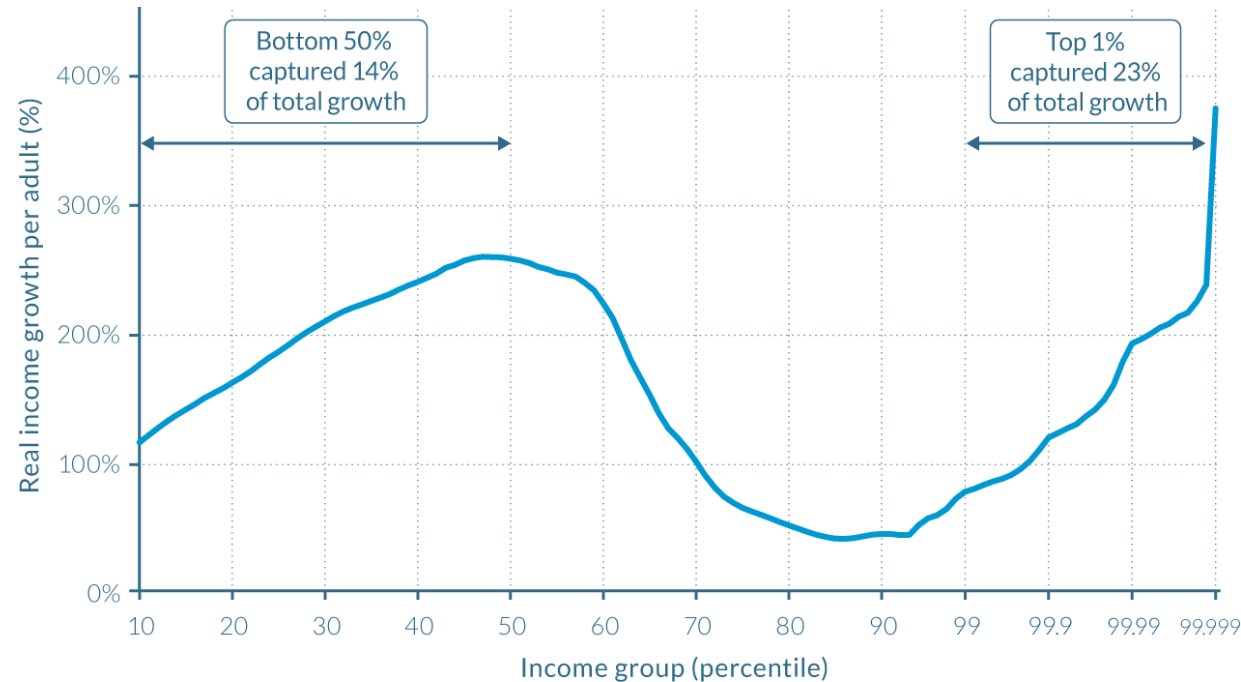
Thyroid Cancer Incidence Rate (age adjusted)
 plotted against glyphosate applied to U.S. corn & soy ($R = 0.988$, $p \leq 7.612e-09$)
 along with %GE corn & soy crops $R = 0.9377$, $p \leq 2.152e-05$
 sources: USDA:NASS; SEER



How to solve this?

*Figure 10, Swanson et al. Journal of Organic Systems 2014; 9(2):6-37.

4. An axis should not change scales midstream



Source: WID.world (2017). See wir2018.wid.world/methodology.html for data series and notes.

On the horizontal axis, the world population is divided into a hundred groups of equal population size and sorted in ascending order from left to right, according to each group's income level. The Top 1% group is divided into ten groups, the richest of these groups is also divided into ten groups, and the very top group is again divided into ten groups of equal population size. The vertical axis shows the total income growth of an average individual in each group between 1980 and 2016. For percentile group p99p99.1 (the poorest 10% among the world's richest 1%), growth was 77% between 1980 and 2016. The Top 1% captured 23% of total growth over this period. Income estimates account for differences in the cost of living between countries. Values are net of inflation.

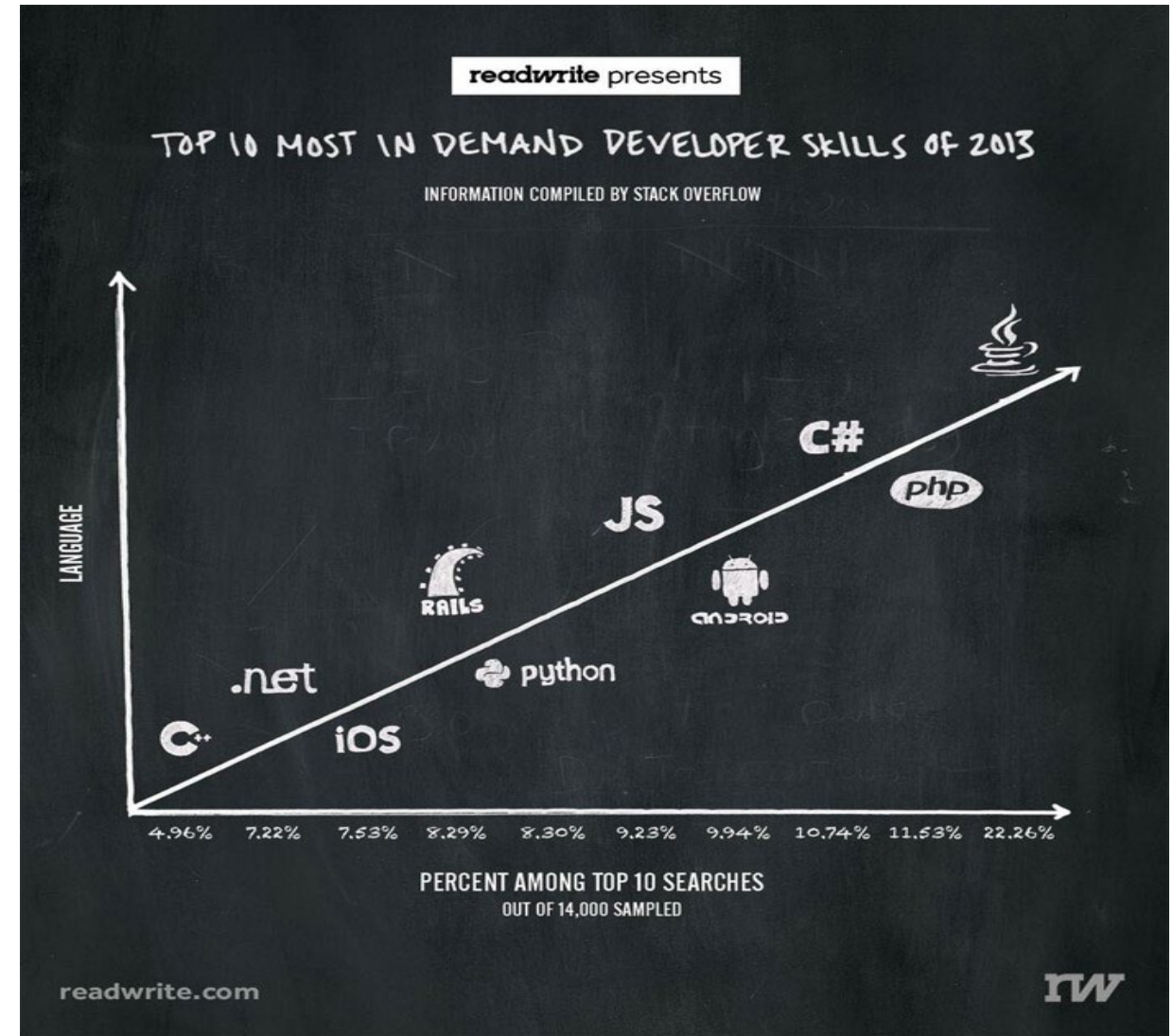
Use log scales!

Is this plot using the “golden ratio”?



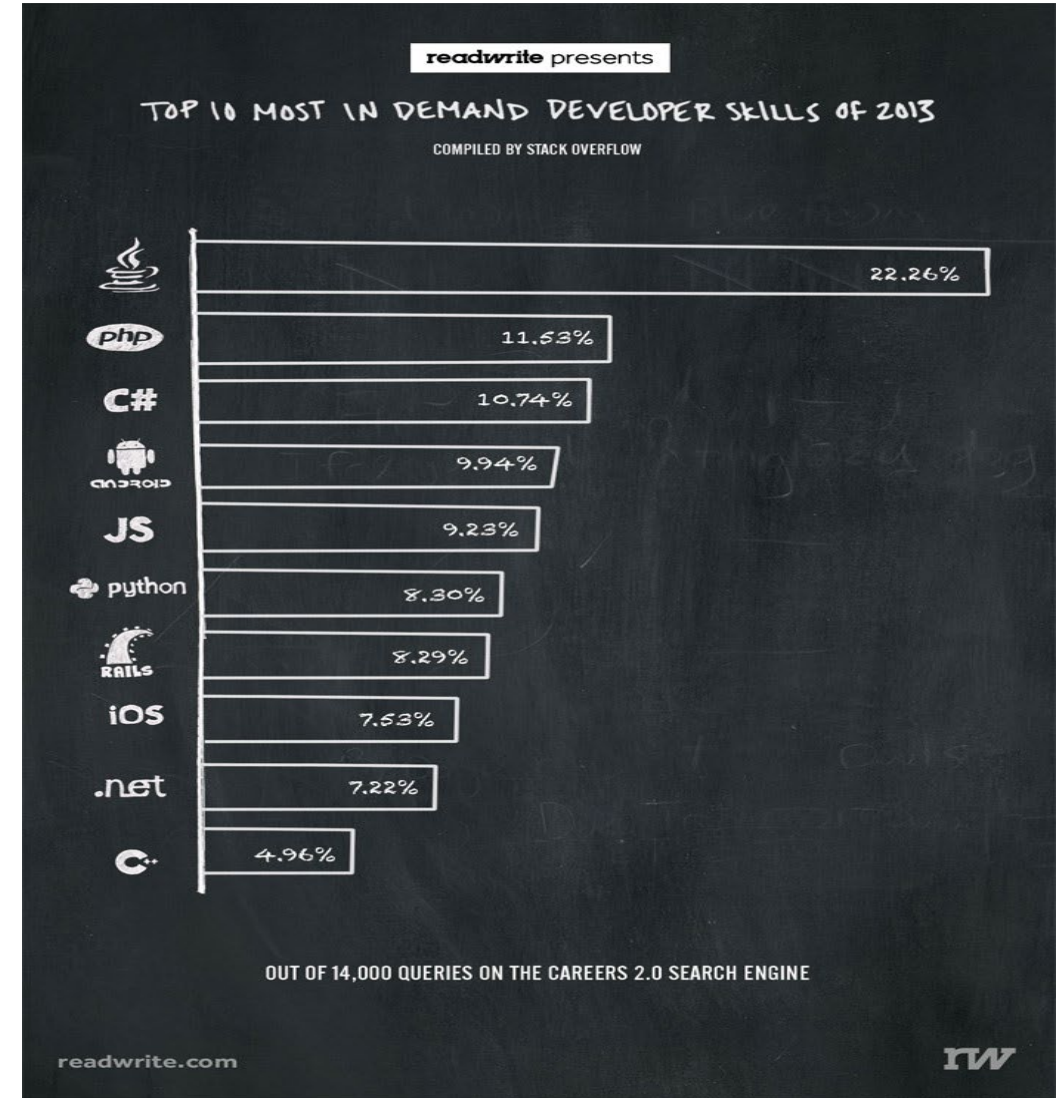


5. An axis should have something on it



Like this

Tools such as
WebPlotDigitizer
can help you get
the missing info

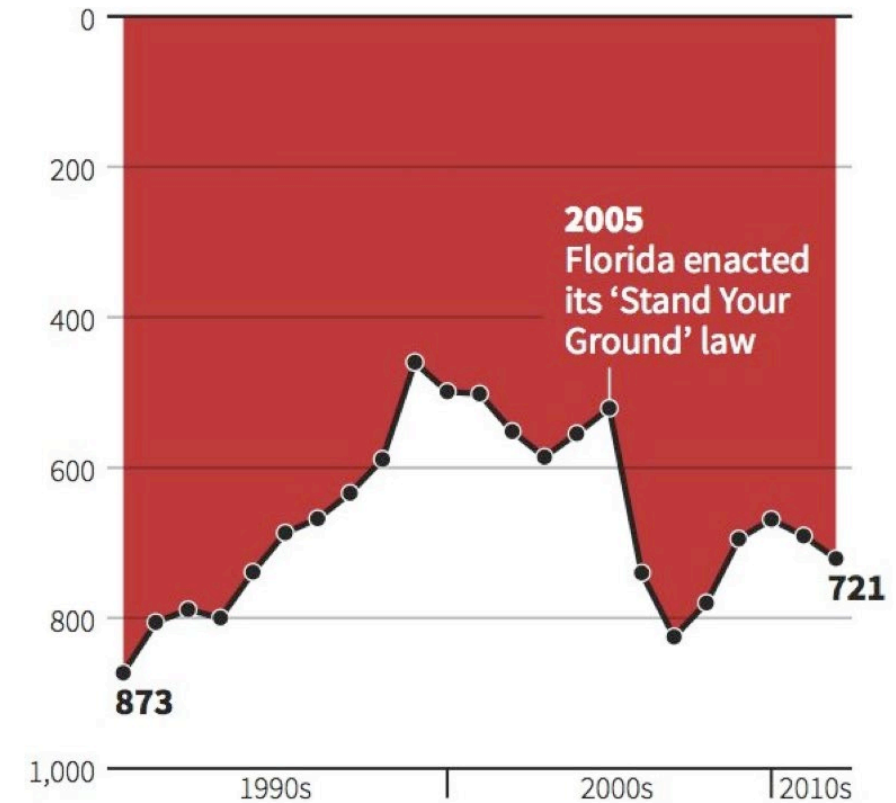


6. Don't invert the axis(?)

The author defended it!
What's your stance?

Gun deaths in Florida

Number of murders committed using firearms



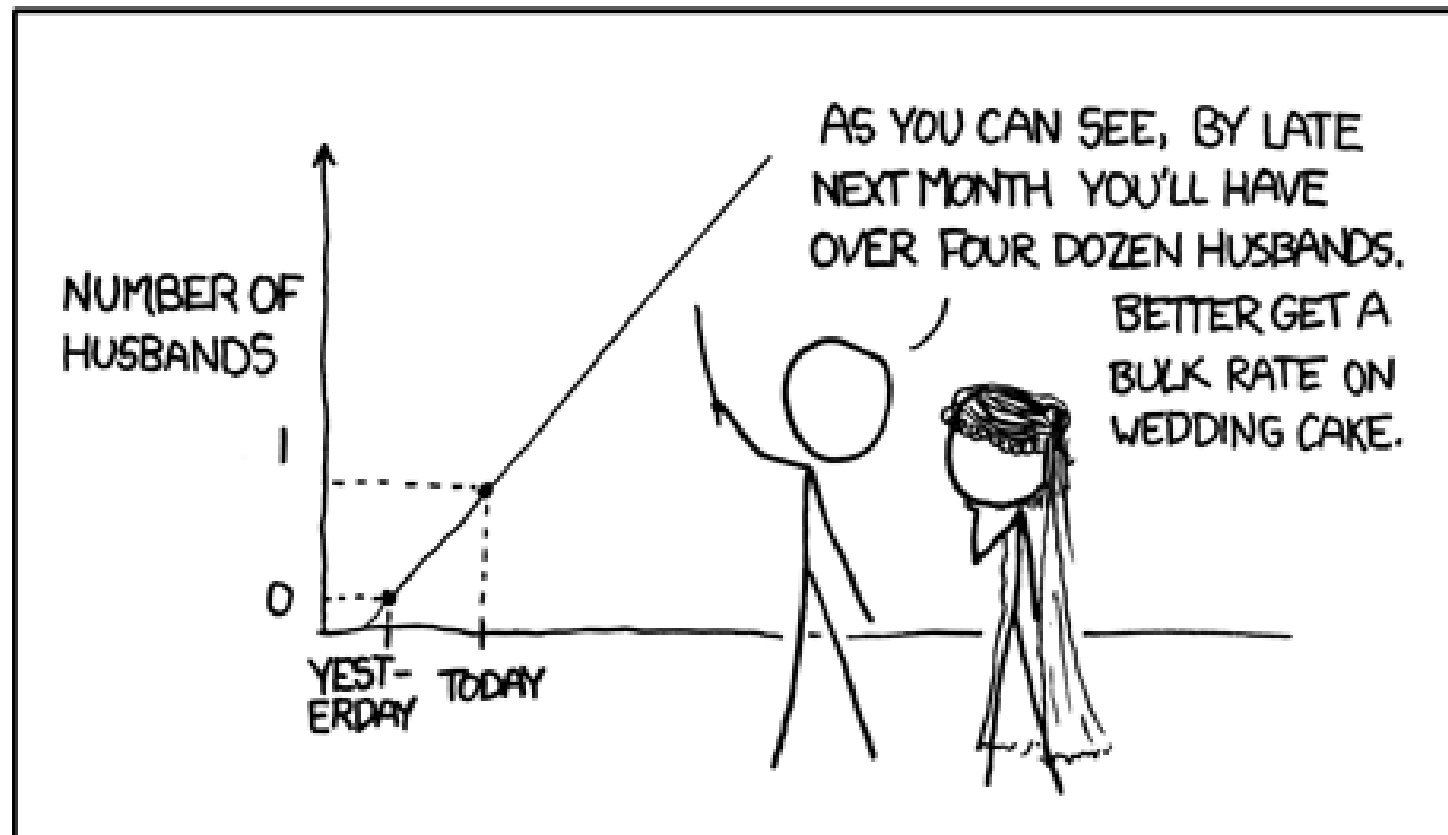
Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

REUTERS

7. Avoid extrapolation!

MY HOBBY: EXTRAPOLATING

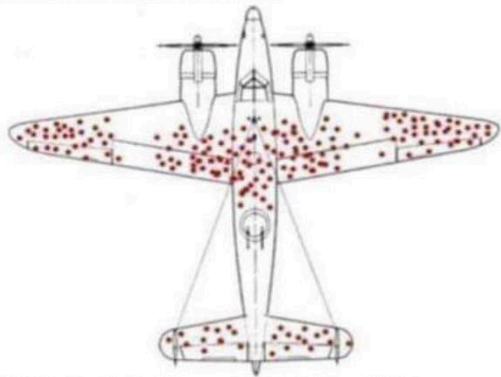


8. Don't jump to conclusions



8. Don't jump to conclusions

This is a picture tracking bullet holes on Allied planes that encountered Nazi anti-aircraft fire in WW2.



At first, the military wanted to reinforce those areas, because obviously that's where the ground crews observed the most damage on returning planes. Until Hungarian-born Jewish mathematician Abraham Wald pointed out that this was the damage on the planes that *made it home*, and the Allies should armor the areas where there are no dots at all, because those are the places where the planes won't survive when hit. This phenomenon is called survivorship bias, a logic error where you focus on things that survived when you should really be looking at things that didn't.



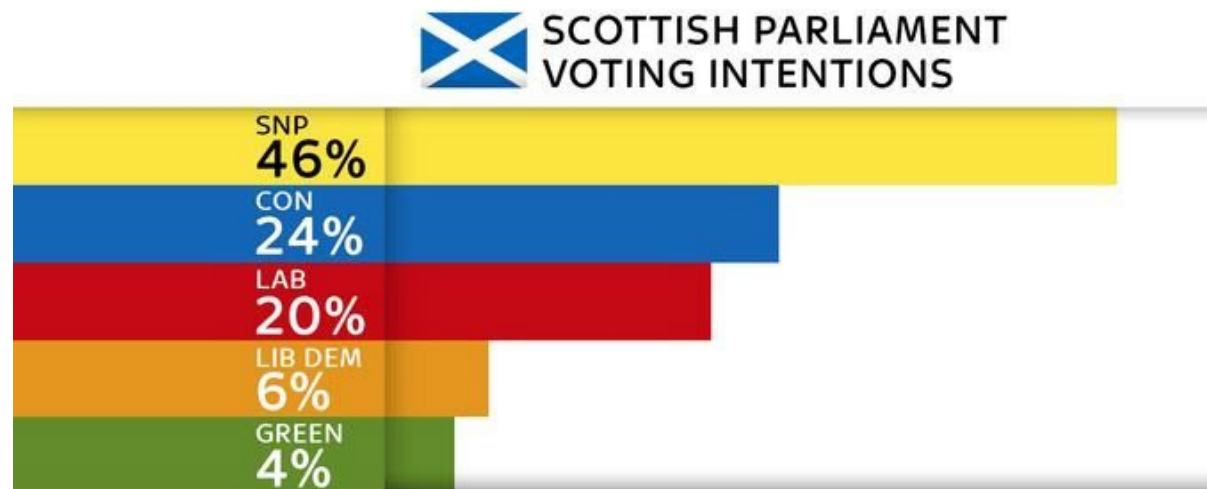
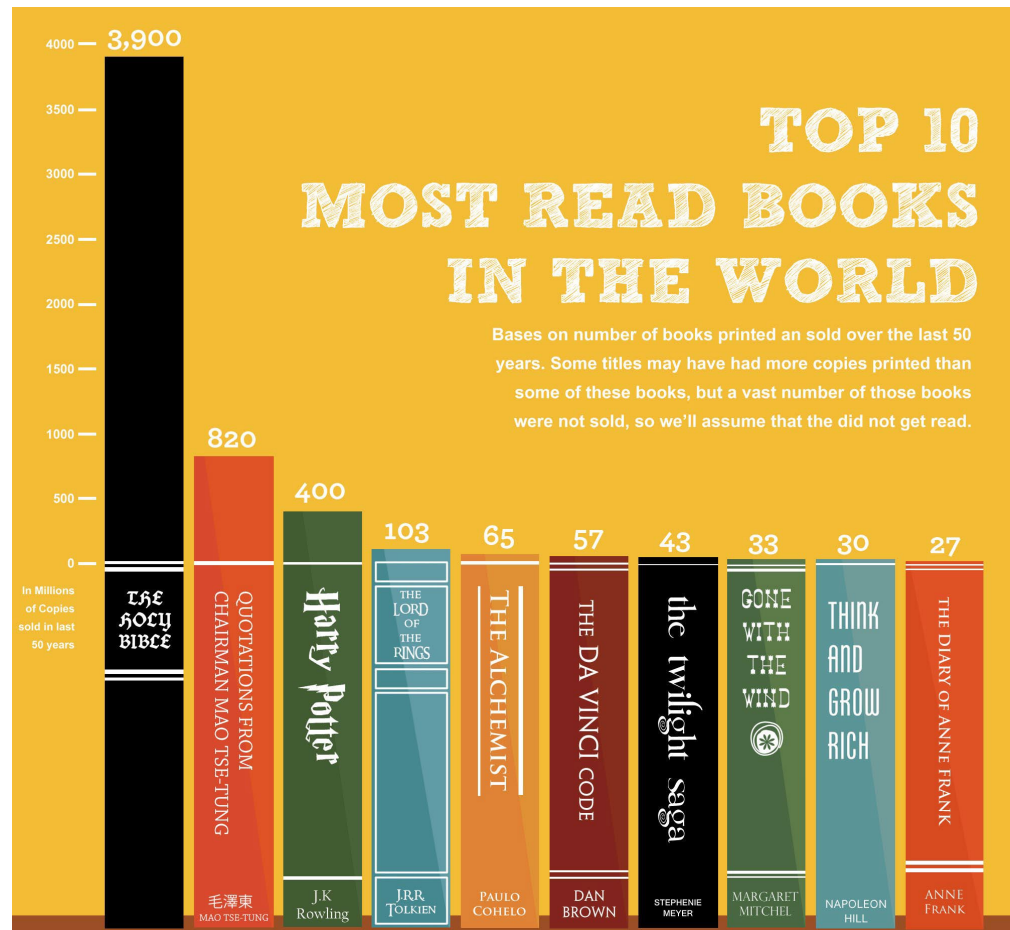
Kirill 00:09

Did you guys know that during the Trojan war the Greeks noticed that their soldiers would return with wounds to the areas around their shield. At first the Greeks wanted to reinforce the areas around their shield until a brilliant mathematician pointed out that it was wounds of soldiers that *MADE IT HOME*, so the Greeks should armor the area where they didn't suffer wounds, ie the shield. (edited)

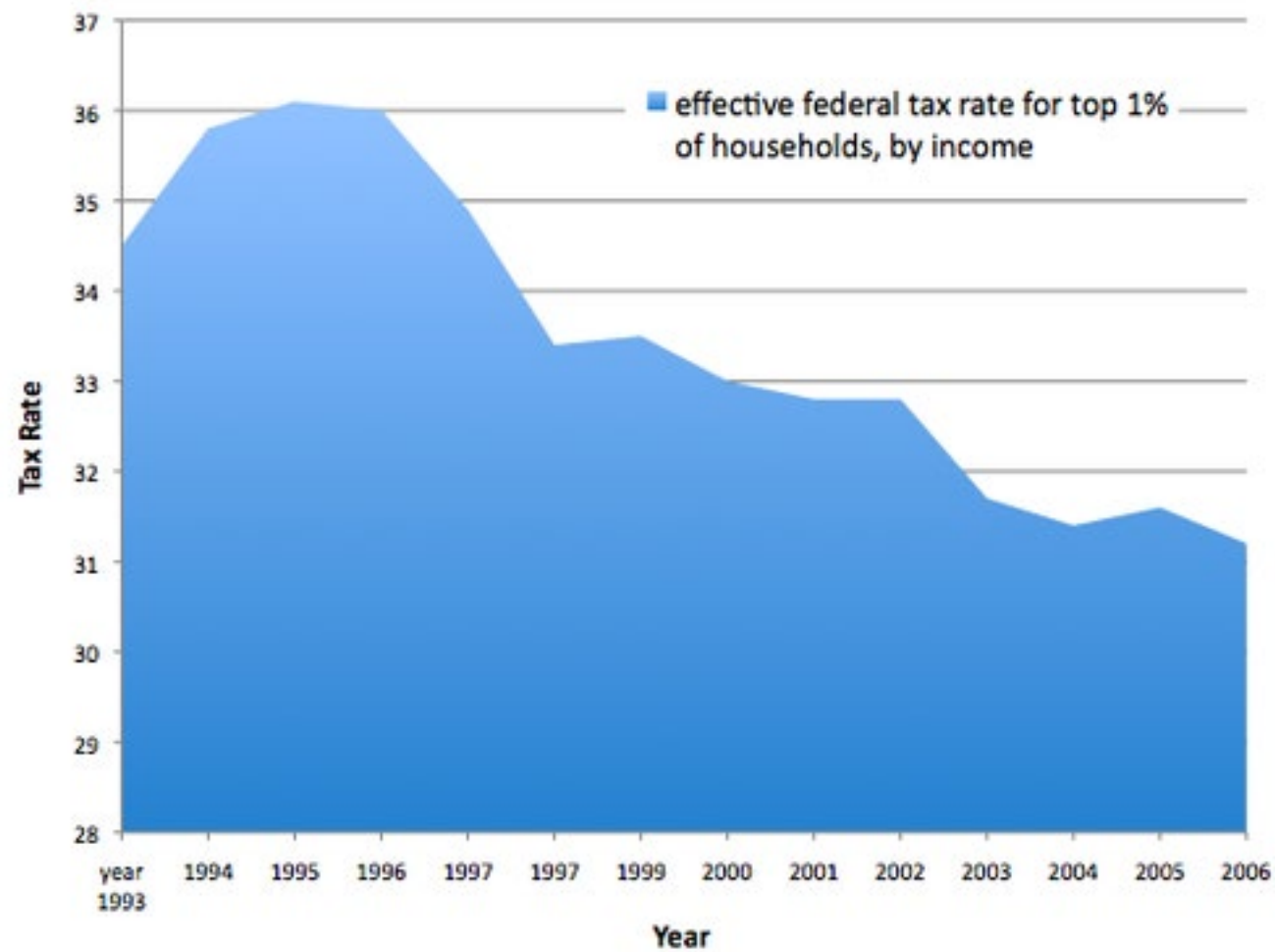


9. Proportional Ink

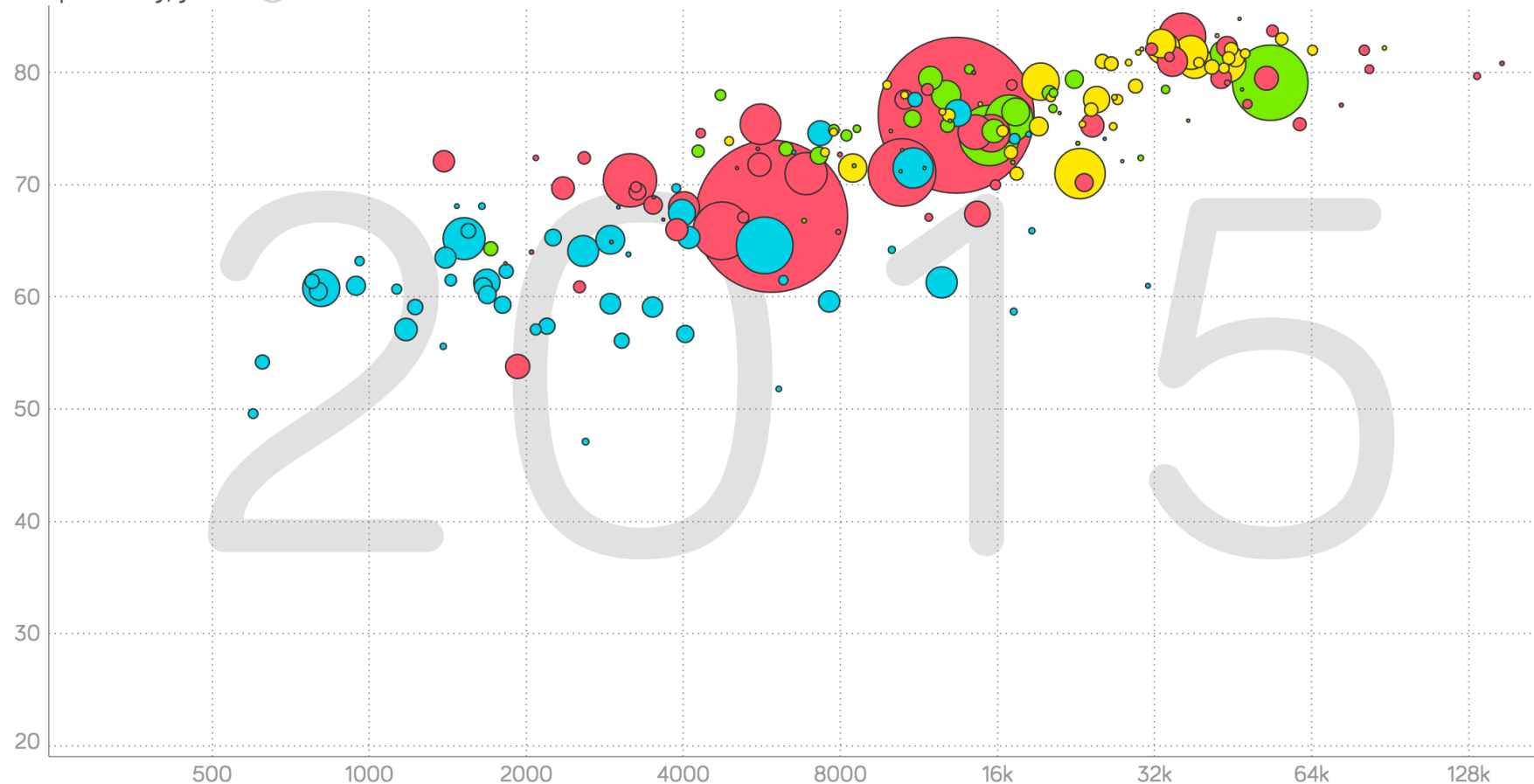
“When a shaded region is used to represent a numerical value, the area of that shaded region should be directly proportional to the corresponding value”



SOURCE: OPINIUM / SKY NEWS



Life expectancy, years ?

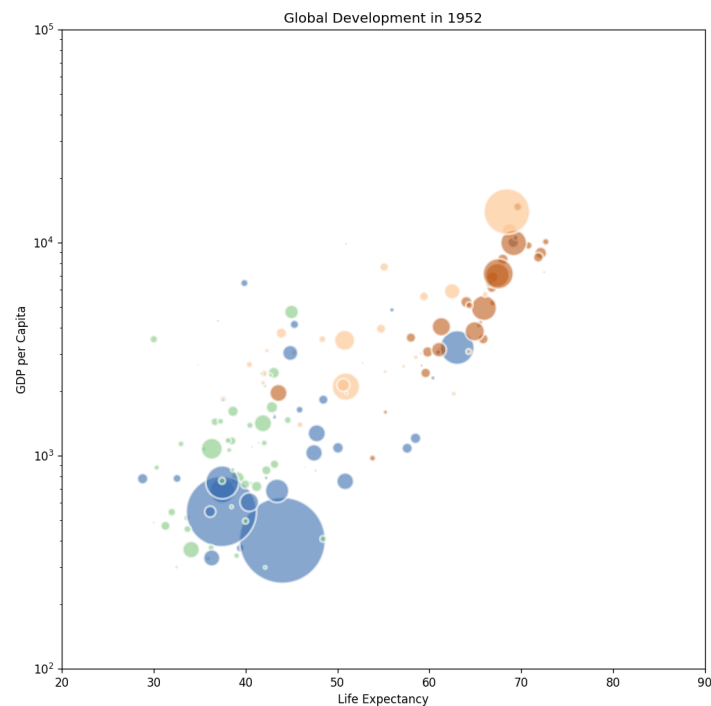


Income per person, GDP/capita in \$/year adjusted for inflation & prices ?

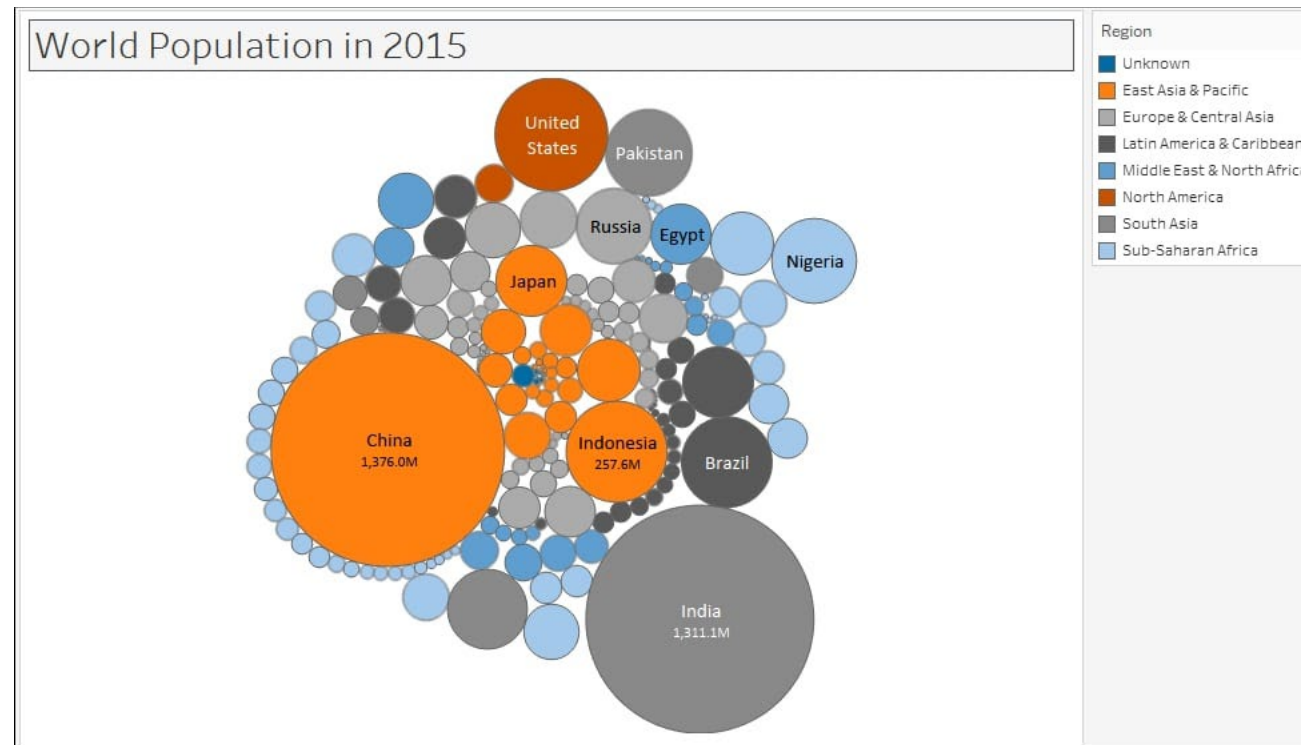
⚠ DATA DOUBTS

Should we violate
the principle?

Some alternatives



<https://python-graph-gallery.com/341-python-gapminder-animation/>



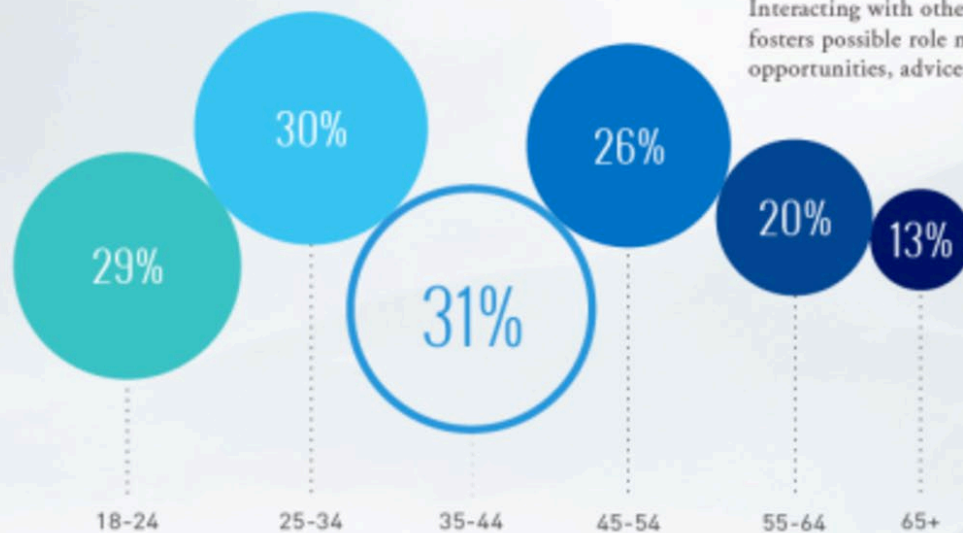
<https://subscription.packtpub.com/book/data/9781786465634/1/ch01/vl1sec12/creating-a-bubble-chart>

Radius $\neq$ Area

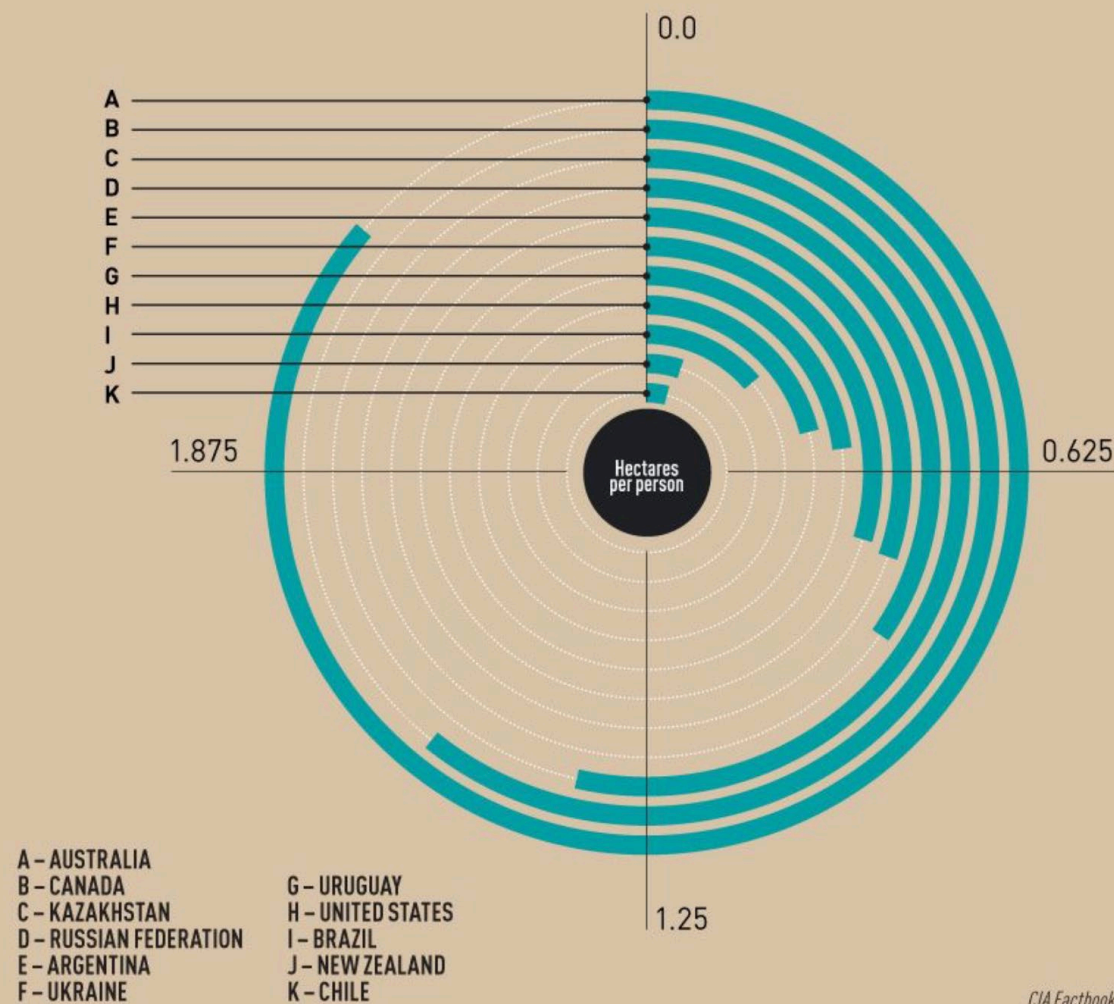
It's Who You Know

By ages 35–44, people are likely to have the strongest network of fellow entrepreneurs, but these **personal social networks** begin to decline over time.

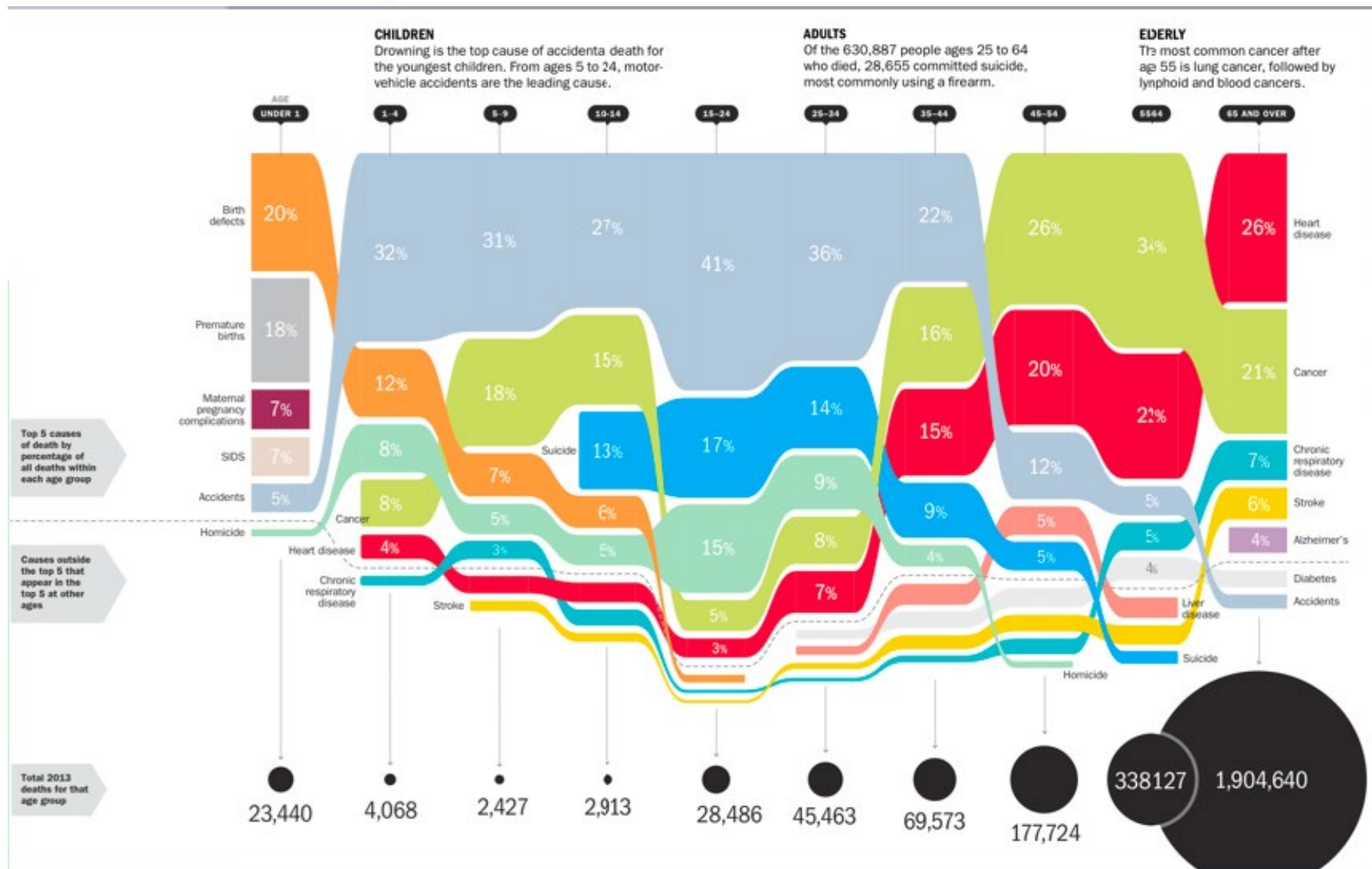
Interacting with other entrepreneurs fosters possible role models, networking opportunities, advice, and encouragement.



ARABLE LAND PER CAPITA (HECTARES PER PERSON)

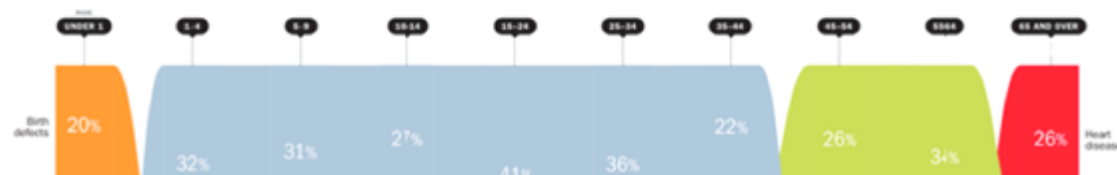


Source:
CIA Factbook 2012 and FAO 2011



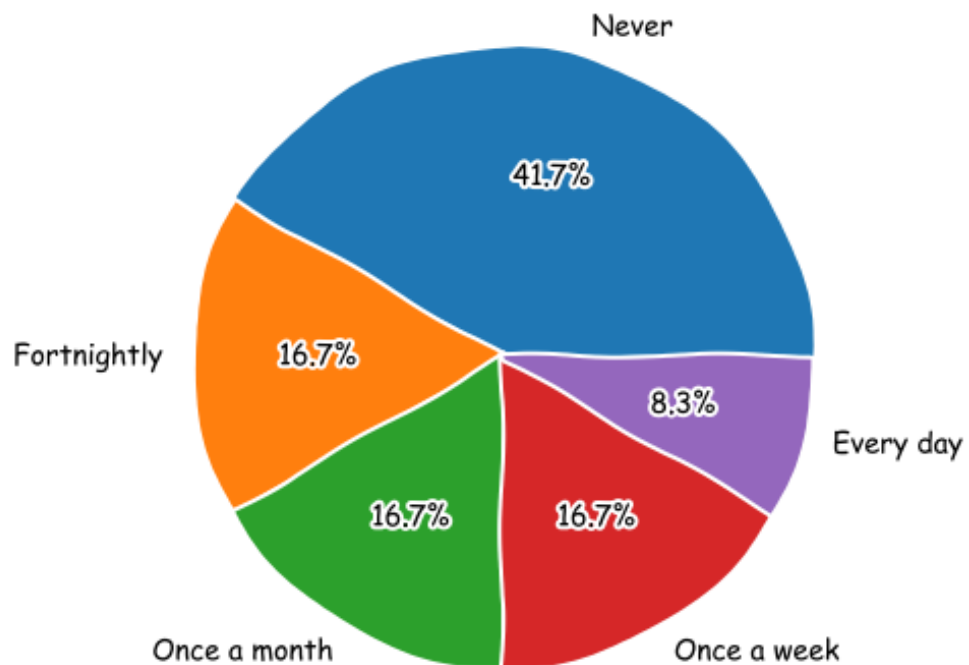
Along vertical slices, ink is proportional to value because shaded areas represent the fraction of a fixed number of deaths (here 28,486) from each cause.

Along horizontal slices, ink is not proportional because total deaths differ widely by age group. Far more people 65 and older died of heart disease (red) than children age 1-4 die of accidents (blue-gray), but the latter takes more ink because it represents a larger percentage of the (relatively few) total deaths at that age.

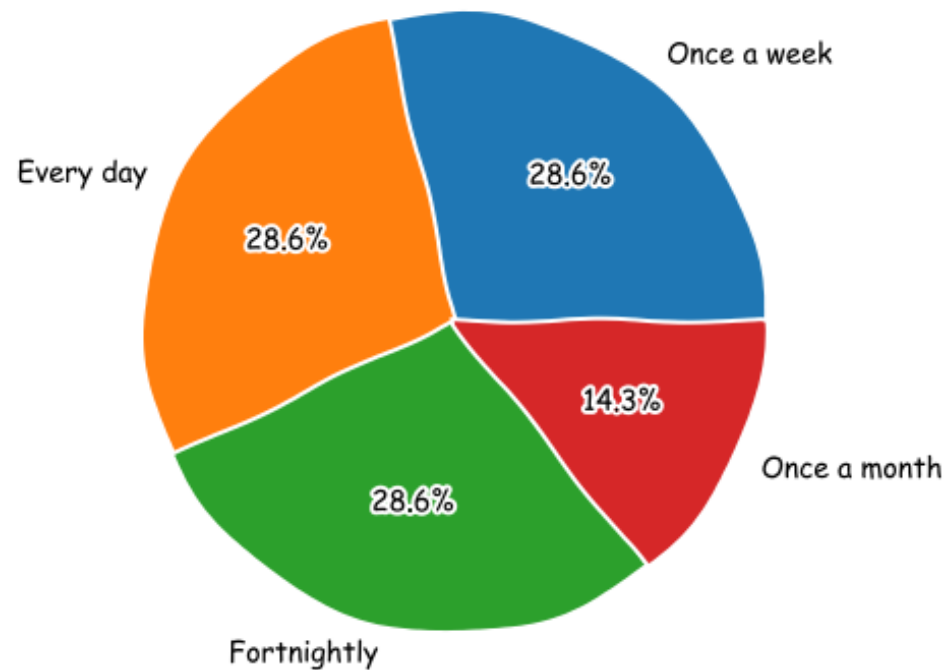


How often do you use a dating app during a month?

Male



Female



**What is wrong in
this one?**

10. Perspective

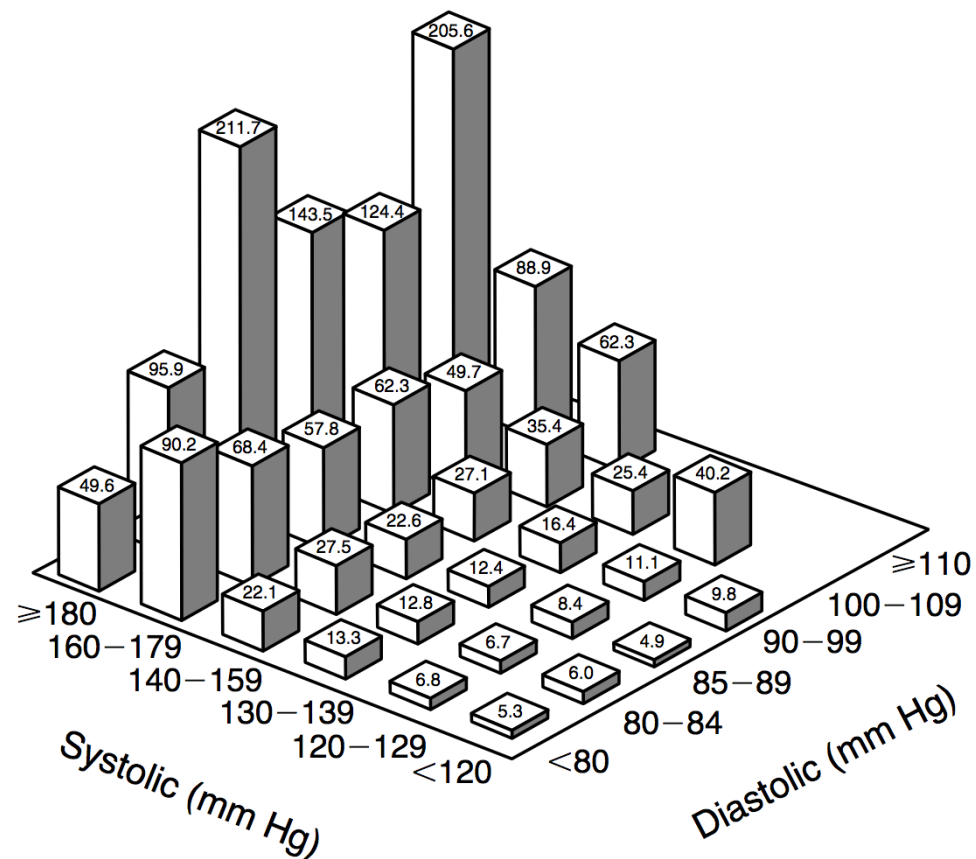
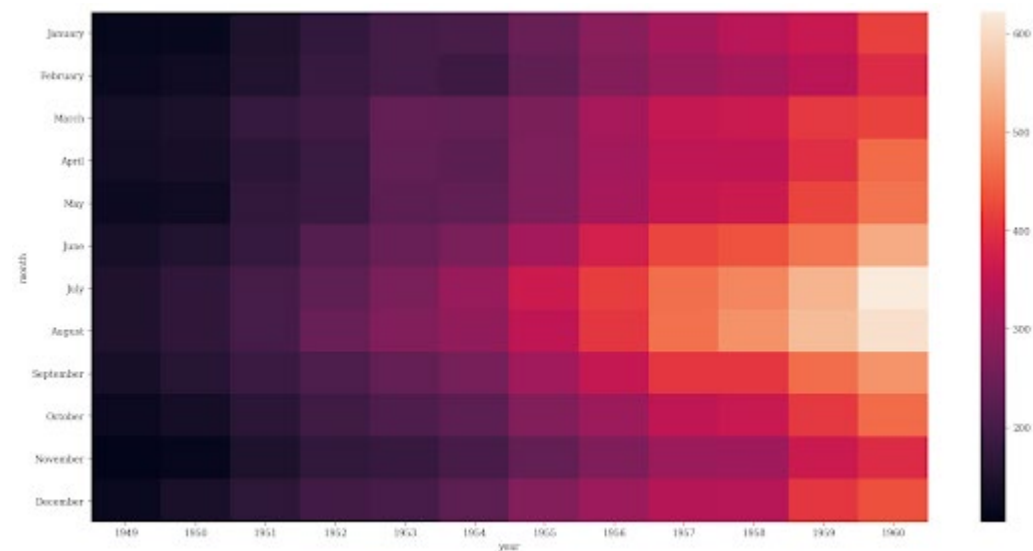
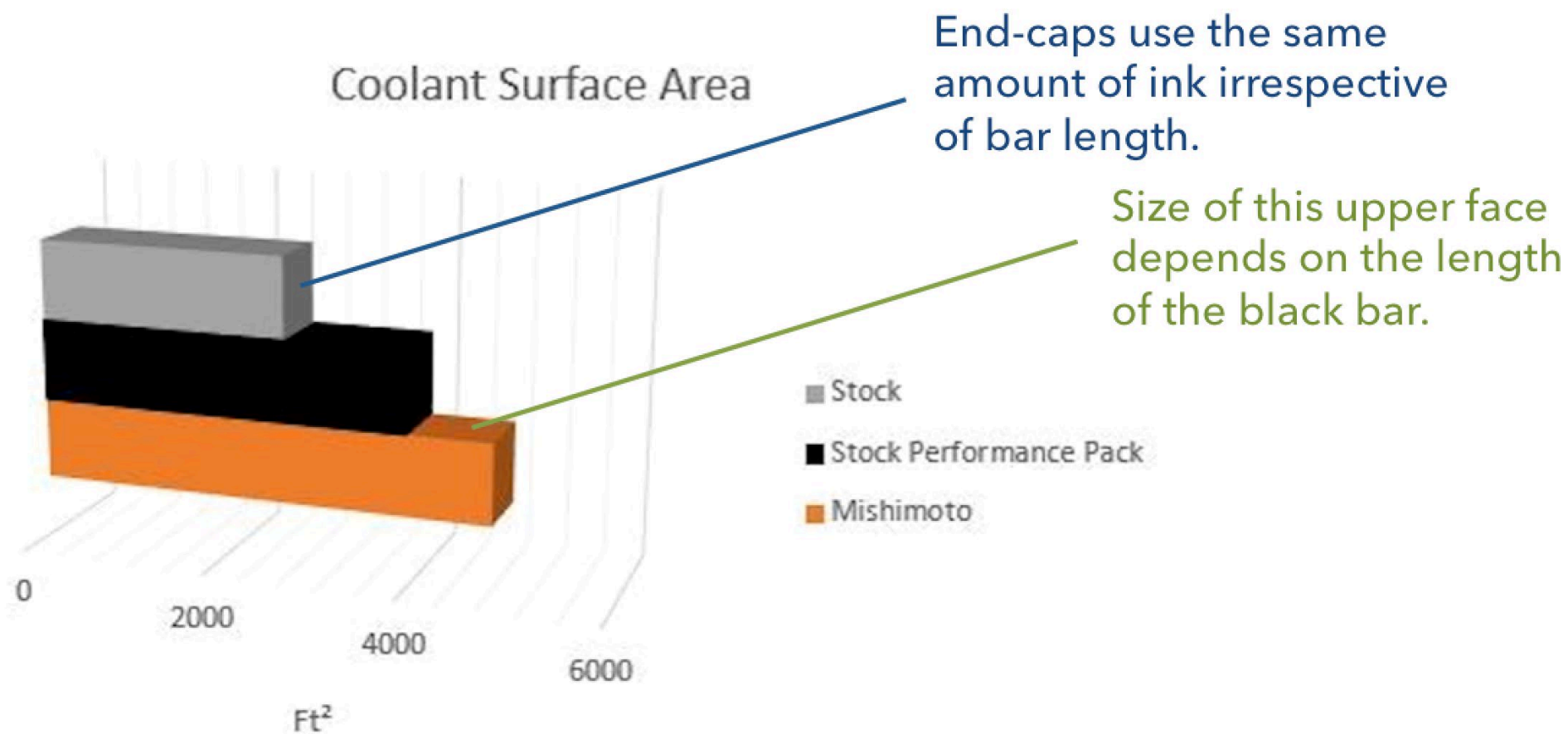
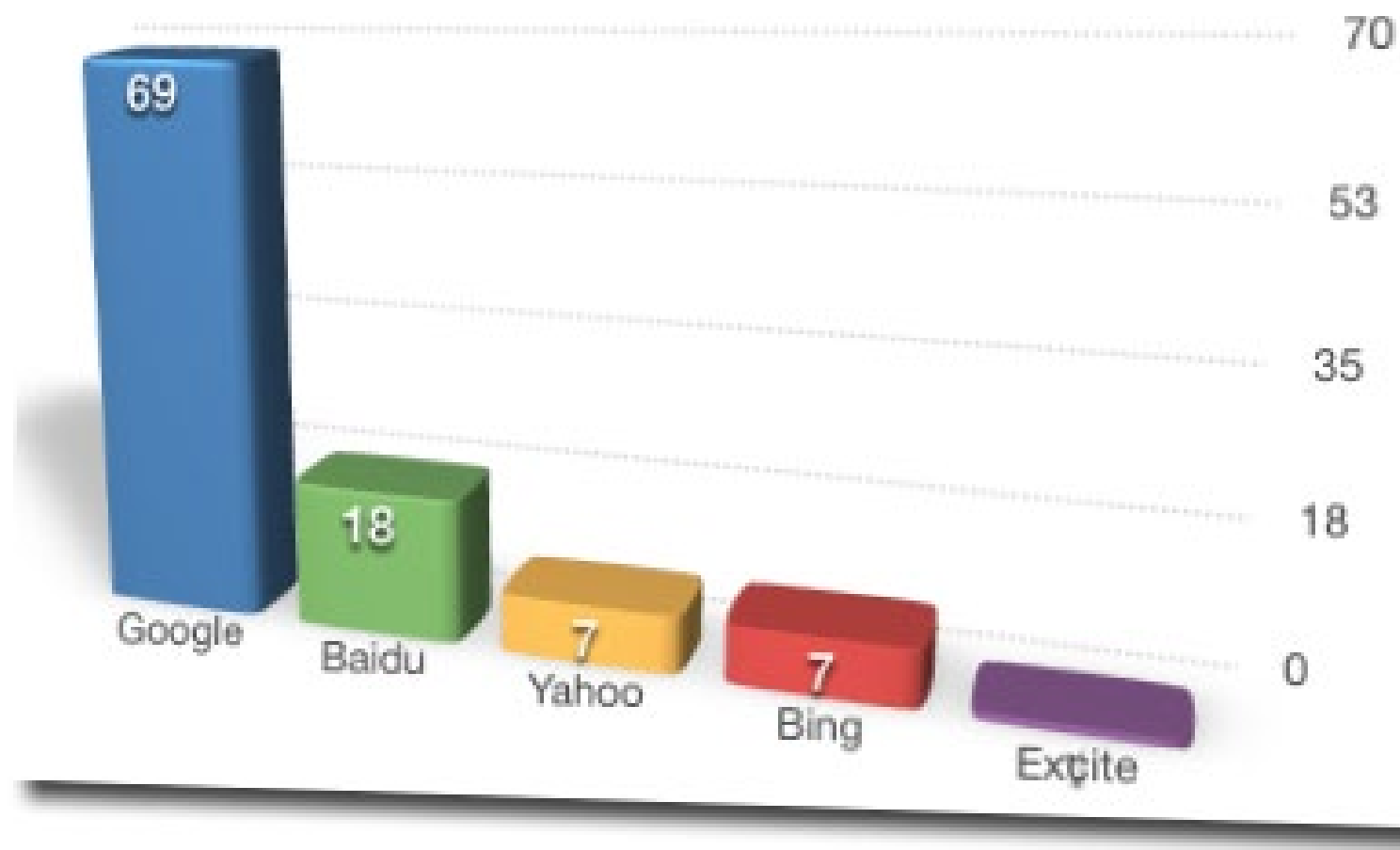


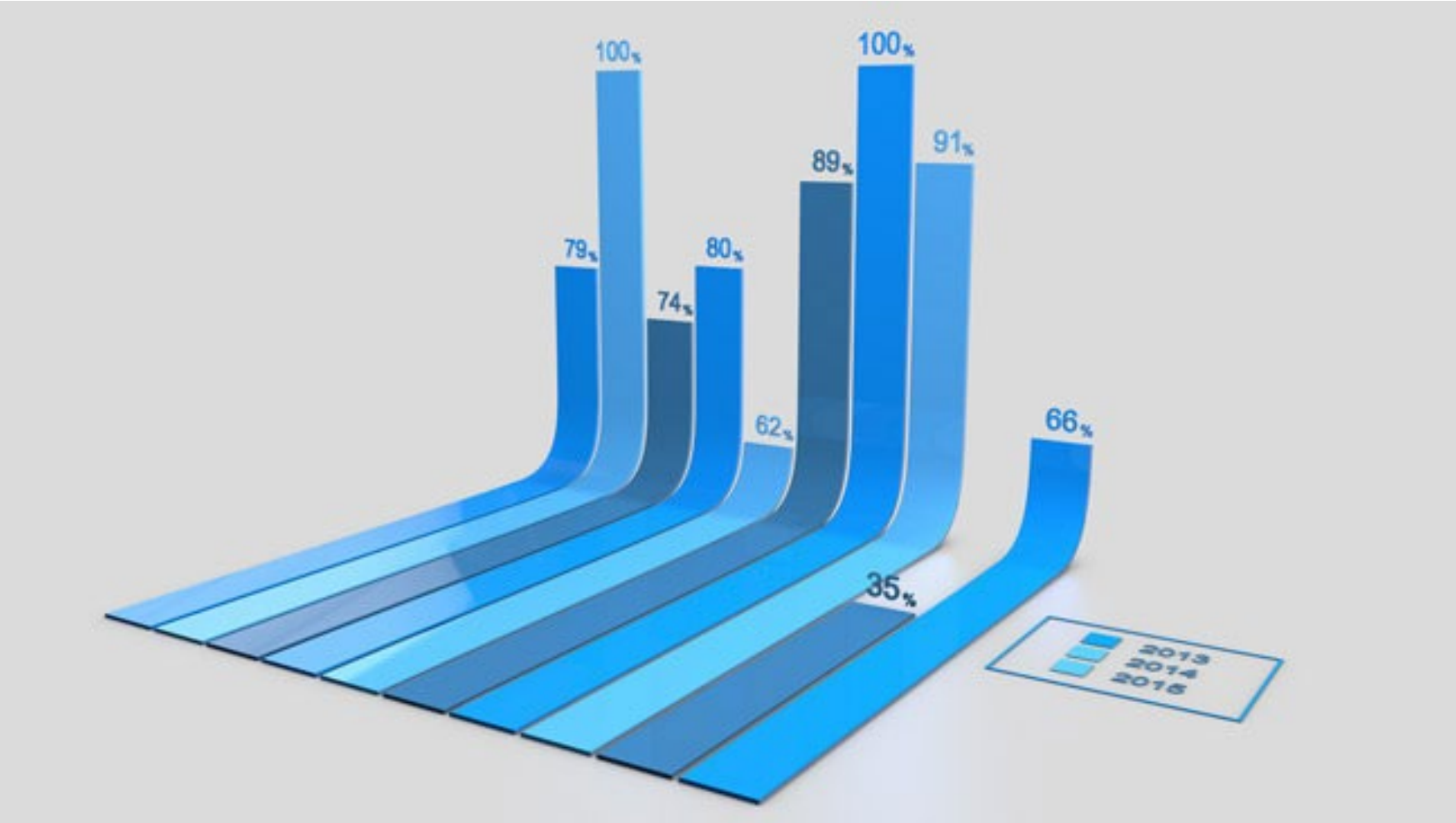
Figure 2. Age-Adjusted Rate of End-Stage Renal Disease Due to Any Cause per 100,000 Person-Years, According to Systolic and Diastolic Blood Pressure in 332,544 Men Screened for MRFIT.



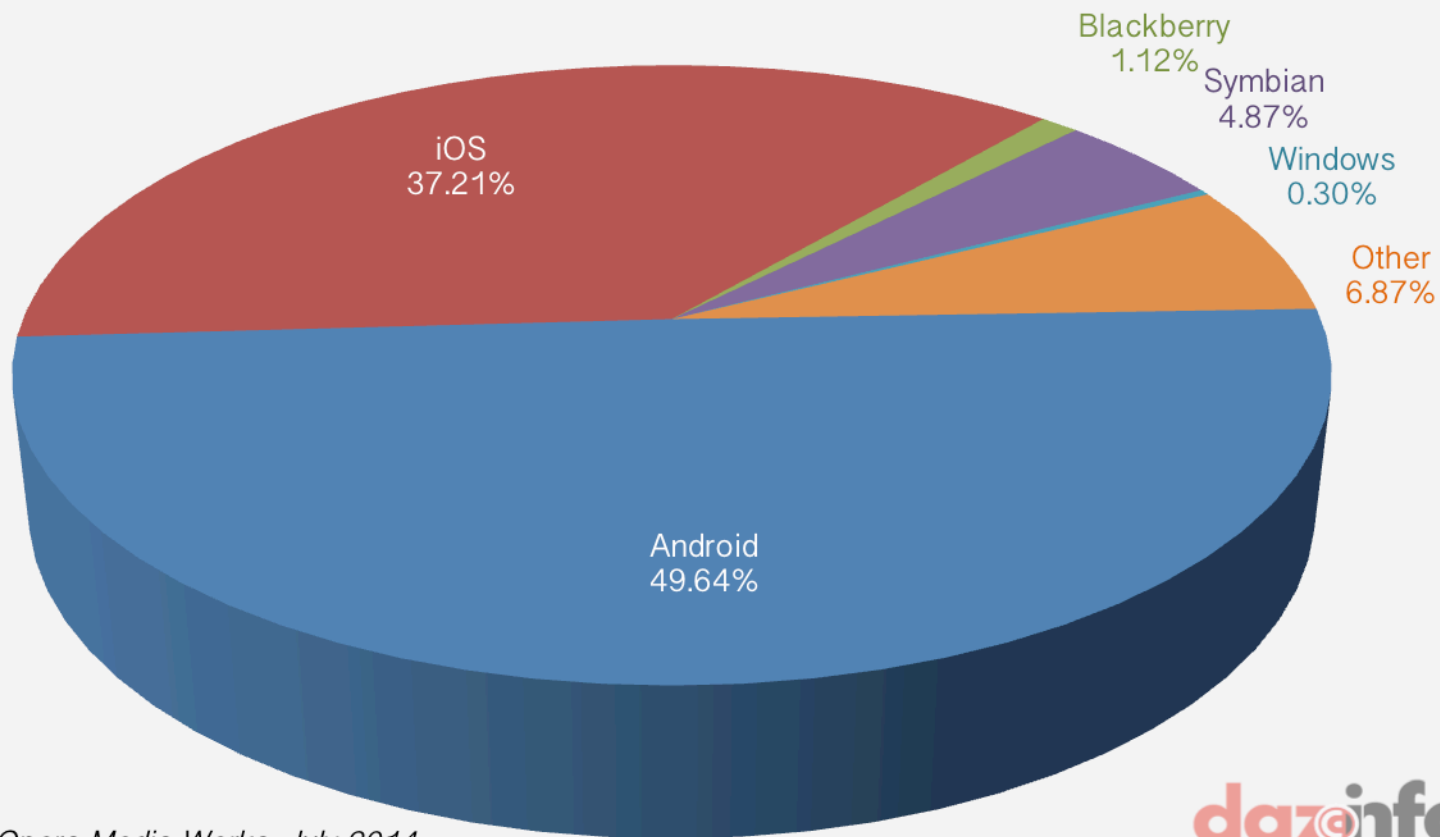


Search Engine Market Share





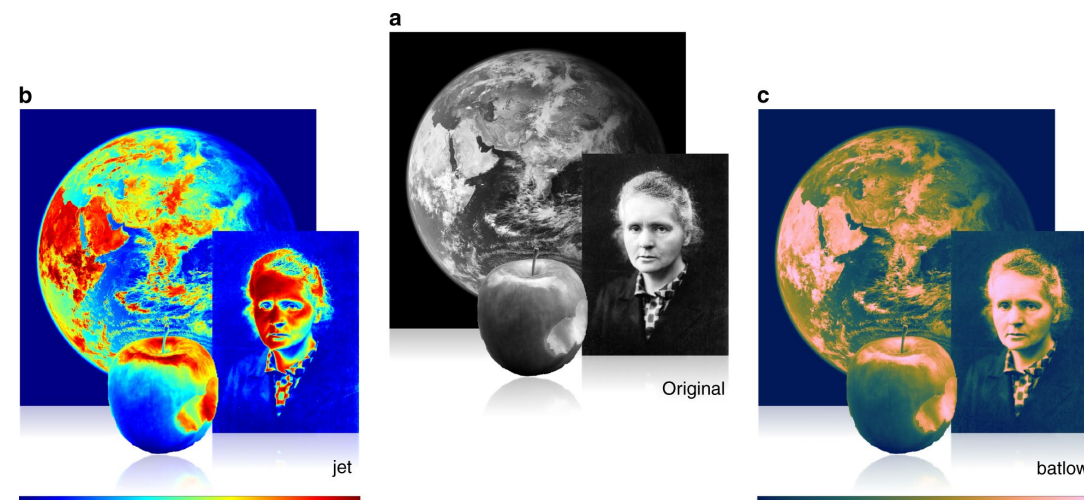
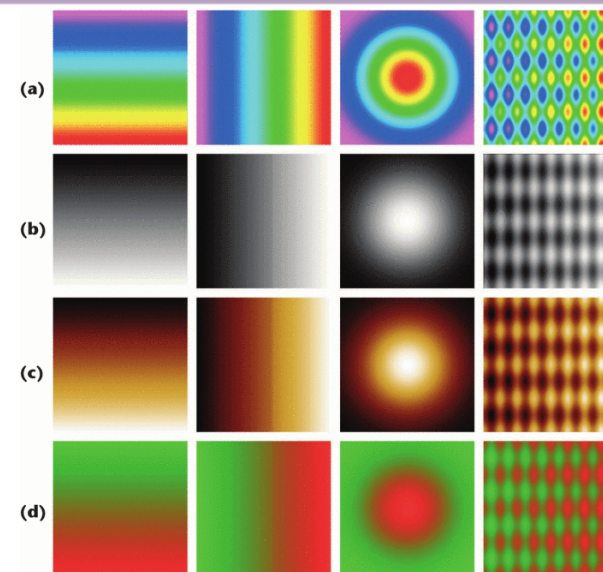
Mobile Phone OS Traffic Share Q2 2014



Source: Opera Media Works, July 2014

11. Colour can lie too!

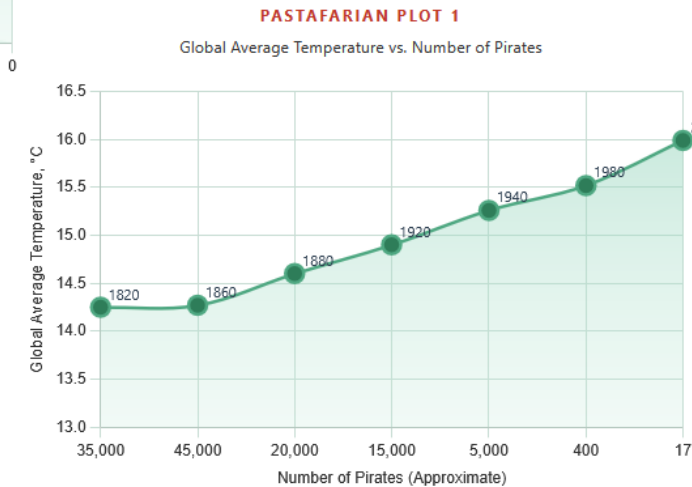
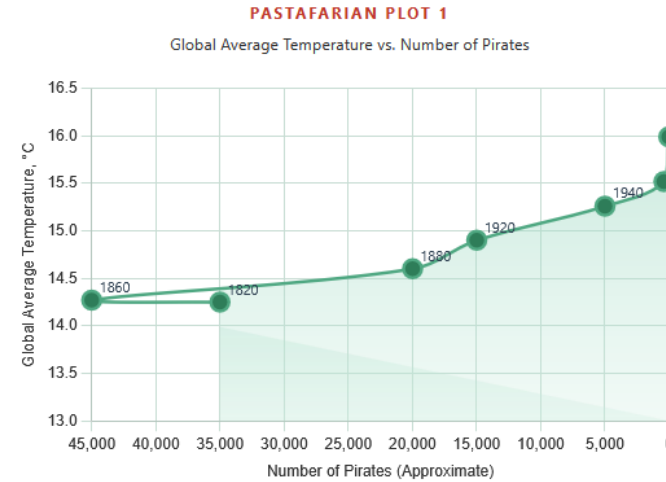
- Rainbow scales make boundaries that aren't in the data
- Uneven colour ramps make some changes look bigger than they are
- Red vs green is indistinct to colour blind people
- If the colour carries the message, ask: how does it look like in greyscale?



D. Borland and R. M. Taylor li, "Rainbow Color Map (Still) Considered Harmful," in IEEE Computer Graphics and Applications, vol. 27, no. 2, pp. 14-17, March-April 2007, <https://doi.org/10.1109/MCG.2007.323435>.
Cramer, F., Shephard, G. E. & Heron, P. J. (2020). The misuse of colour in science communication. *Nature Communications*, 11, 5444.
<https://www.nature.com/articles/s41467-020-19160-7>

What happens when machines draw the charts?

- An AI can produce a chart and a caption in seconds!
- However, it could break the rules and even make some new mistakes.
- A human who misleads usually knows it.
- A model doesn't, and it sounds just as sure either way.



Five common mistakes

1. **Values that were never there:** gaps filled by the model
2. **The popular chart, not the right one:** picks what it has seen most, not what fits
3. **Causal captions:** "X drives Y" from a correlation
4. **Correct code, meaningless numbers:** mean of ratios, rows double-counted after a join
5. **Sources that don't exist;** invented citations and dead links

Incorrectly calculated values

skip_rate.py

df.head()

month	plays	skips	skip_rate
Jan	20	12	0.600
Feb	450	45	0.100
Mar	380	38	0.100

what the AI wrote

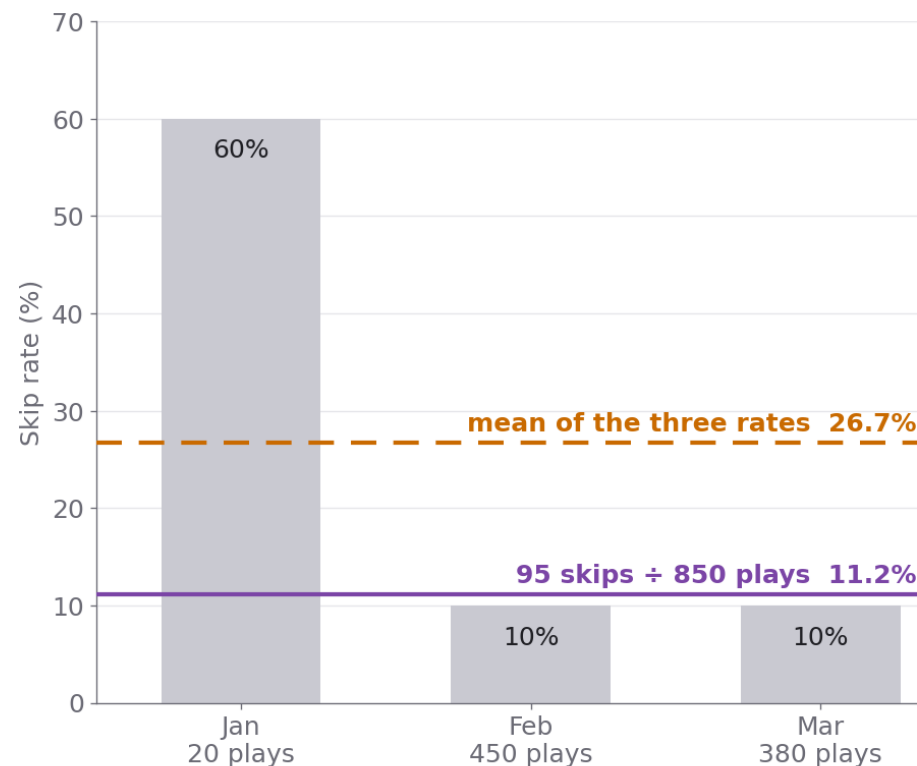
df['skip_rate'].mean()

→ 26.7%

what it should be

df['skips'].sum() / df['plays'].sum()

→ 11.2%



Incorrect Data Joins

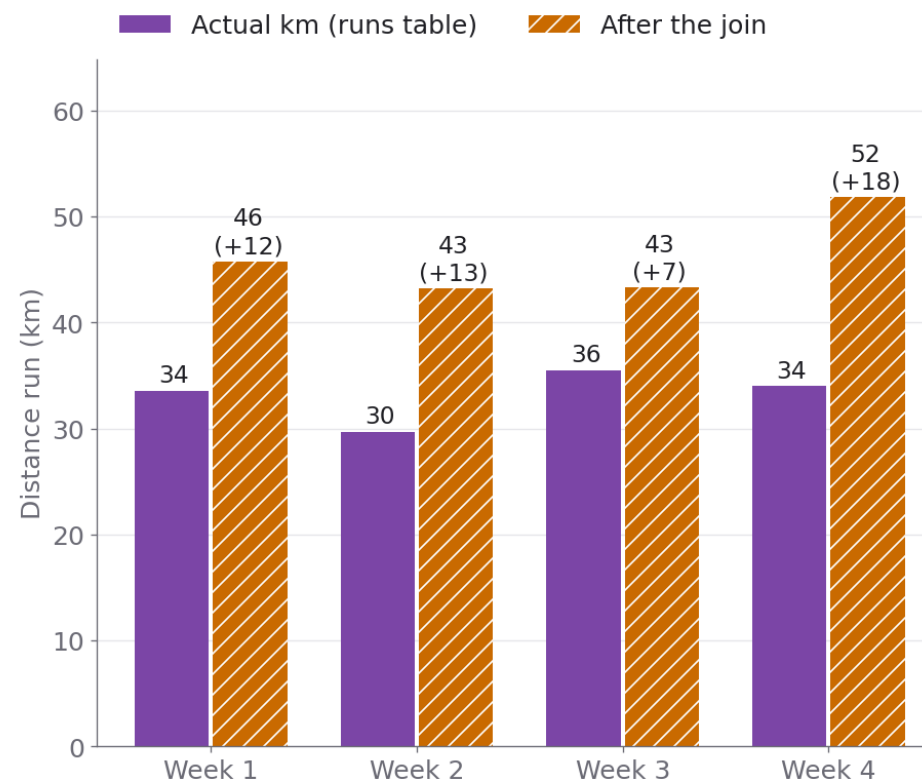
```

weekly_km.py

runs.shape      # one row per run
→ (16, 2)
weather.shape   # AM + some PM readings
→ (35, 3)

# what the AI wrote
df = runs.merge(weather, on='date')
df.shape → (23, 4)
df['km'].sum() → 184.2 km

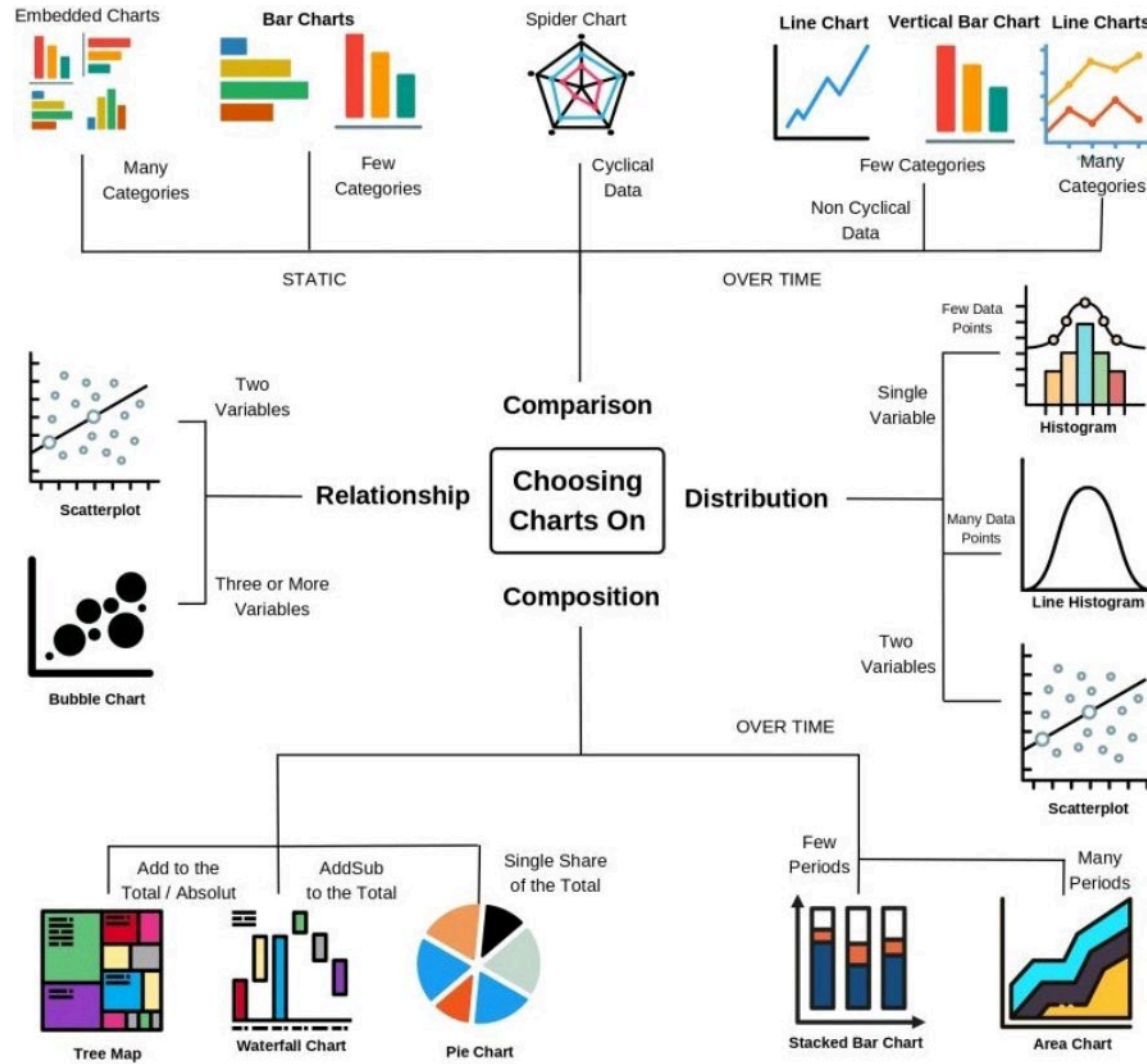
# what it should be
daily = weather.groupby('date')['temp_c'].mean()
runs.merge(daily, on='date', validate='1:1')
→ 132.8 km
    
```



Can AI call bullsh*t for you?

- Well, the previous two examples were generated by AI (in case you didn't know already) so technically, it can!
- Researchers like Lo et al. have tested multimodal LLMs on misleading charts
- Results depended heavily on how the question was asked and the pre-existing knowledge of the model
- AI is useful as a second reader, but not as the final judge

So, is there a way to choose the “right” plot?

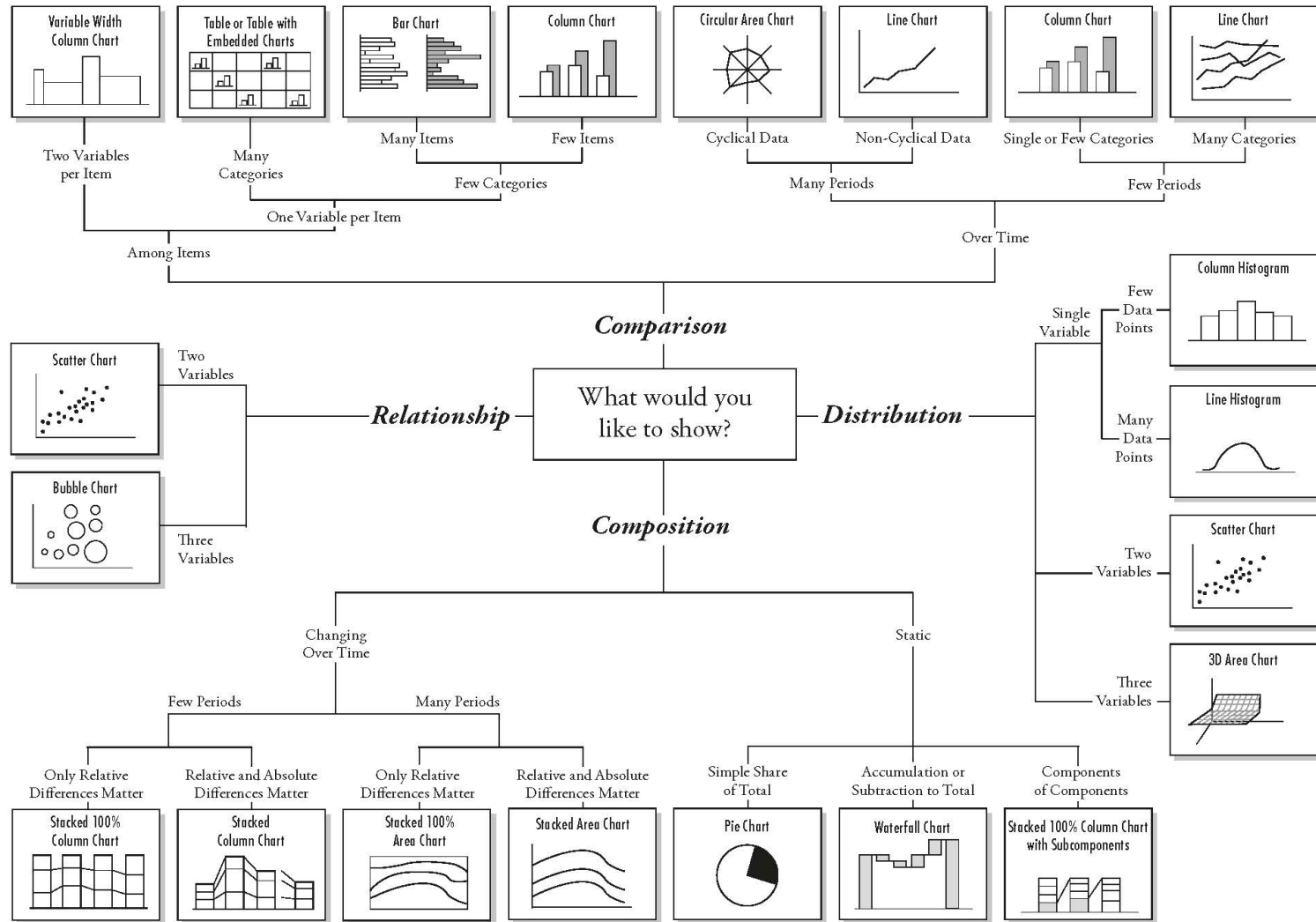


By Johnny Shollaj

Add me on:



Chart Suggestions—A Thought-Starter



Deviation

Emphasise variations (+/-) from a fixed reference point. Typically the reference point is zero but it can also be a target or a long-term average. Can also be used to show sentiment (positive/negative).

Example FT uses
Trade surplus/deficit, climate change

Diverging bar
A simple standard bar chart that can handle both negative and positive magnitude values.

Diverging stacked bar
Perfect for presenting binary results which involve sentiment (eg disagreement/agree).

Spine
Splits a single value into two contrasting components (eg market/share).

Surplus/deficit filled bar
The filled area of these charts allows a balance to be shown – either against a baseline or between two series.

Correlation

Show the relationship between two or more variables. Be mindful that, unless you tell them otherwise, many readers will assume the relationship you show *must* be causal (one caused the other).

Example FT uses
Inflation and unemployment, income and life expectancy

Scatterplot
The standard way to show the relationship between two continuous variables, each of which has its own axis.

Column + line timeline
A good way of showing the relationship between an amount (columns) and a rate (line).

Connected scatterplot
Usually used to show how the relationship between 2 variables has changed over time.

Bubble
Like a scatterplot, but with additional data by using the circles according to a third variable.

XY heatmap
A good way of showing the pattern between 2 categories of data, less effective at showing fine differences in amounts.

Ranking

Use where an item's position in an ordered list is more important than its absolute or relative value. Don't be afraid to highlight the points of interest.

Example FT uses
wealth, deprivation, league tables, constituency election results

Ordered bar
Standard bar charts display the values of values much more easily when ranked in order.

Ordered column
See above.

Ordered proportional column
Use when there are big variations between components. Can be a problem when too many data have the same value.

Dot strip plot
Like dot strip plots, good for displaying all the data in a table. They work best when points without missing a key part of story.

Shape
Perfect for showing how rates have changed over time or vary between categories.

Lollipop
Lollipop draw more attention to the data value than ordered bar charts and can also show rate and value effectively.

Bump
Effective for showing changes in rankings. For larger datasets, consider grouping lines using colour.

Distribution

Show values in a dataset and how often they occur. The shape of the distribution can be a noticeable way of highlighting the lack of uniformity or equality in the data.

Example FT uses
income distribution, population, expenditure distribution, revealing inequality

Histogram
The standard way to show a statistical distribution. Keep the gaps between columns small to highlight the shape of the data.

Dot plot
A simple way of showing the shape or range (minimums of data across multiple categories).

Dot strip plot
Good for showing individual values in a distribution. Can be a problem when too many data have the same value.

Barcode plot
Good for displaying all the data in a table. They work best when points without missing a key part of story.

Boxplot
Summarise multiple distributions by showing the median (central) and range of the data.

Violin plot
Similar to a box plot but more effective with complex distributions (data that cannot be summarised with simple averages).

Population pyramid
A standard way for showing the age and sex distribution of a population distribution. Effective, back to back histograms.

Cumulative curve
A good way of showing how cumulative a distribution is. It's axis is always cumulative frequency, axis is always a measure.

Frequency polygons
For displaying multiple distributions of data. Like a regular line chart, best suited to a maximum of 3 or 4 datasets.

Beeswarm
Use to emphasise individual points in a distribution. Points can be sized to an additional variable. Best with medium-sized datasets.

Change over Time

Give emphasis to changing trends. These can be short (one day) movements or extended series, however decades or centuries. Choosing the correct time period is important to provide subtle context for the reader.

Example FT uses
Share price movements, economic time series, sectoral changes in a market

Line
The standard way to compare the value of things that change over time. But be aware that it's difficult to see small differences in the data at a time.

Column
Columns work well for showing change over time – but usually best with only one series and labels that are long or many categories.

Column + line timeline
A good way of showing the relationship over time. Includes ticks for the date and a rate (line).

Slope
Good for showing changing data as long as the data can be simplified into 2 or 3 key parts of story.

Area chart
Use with care – these are good at showing changes in total, but seeing change in the data can be very difficult.

Candlestick
Usually focused on day-to-day activity, they clearly show opening/closing and high/low points of each day.

Fan chart (projection)
Use to show the uncertainty in future projections. Usually the further forward to projection.

Connected scatterplot
A good way of showing changing data for two variables where there is a relatively clear pattern of progression.

Calendar heatmap
A great way of showing temporal patterns. (Daily, weekly, monthly) – at the expense of showing precision in quantity.

Priority timeline
Great when date and duration are key elements of the story in the data.

Circle timeline
Good for showing a sequence of events or varying size across multiple categories (eg earthquakes by magnitude).

Vertical timeline
Presents time on the Y axis. Good for displaying detailed time series that work especially well when scrolling on mobile.

Solenogram
Another alternative to the circle timeline for priority series where there are big variations in the data.

Streamgraph
A type of area chart, use when seeing changes in proportions over time is more important than individual values.

Magnitude

Show size comparisons. These can be relative (can be being able to see larger/smaller) or absolute (need to see the difference). Usually these show a 'baseline' number (for example, barrels, dollars or pounds) rather than a calculated rate or per cent.

Example FT uses
Commodity production, market capitalisation, volumes in general

Column
The standard way to compare the value of things. Must always start at 0 on the axis.

Bar
See above. Good when the data are not time series and labels that are long or many categories.

Paired column
As per standard column but allows for multiple series. Can include ticks for the date and a rate (line).

Paired bar
See above.

Horizontal
A good way of showing the size and proportion of data at the same time – as long as the data are not too complicated.

Proportional symbol
Use when there are big variations between values and the scale is close to the central point than any other context.

Isotype (pictograph)
Excellent solution in use for illustrating comparative composition by number of items (a decimal).

Lollipop
Lollipop charts draw more attention to the data value than standard bar charts – does not have to start at zero (but preferred).

Radar
A space-efficient way of showing value of multiple variables – but make sure they are organised in a way that makes sense to reader.

Parallel coordinates
An alternative to radar charts – again, the arrangement of the variables is important. Usually benefits from highlighting values.

Butter
Good for showing a measure across a target or performance range.

Grouped symbol
An alternative to bar charts when trying to compare data in groups. Individual elements are visible.

Part-to-whole

Show how a single entity can be broken down into its component elements. If the reader's interest is solely in the size of the components, consider a magnitude-type chart instead.

Example FT uses
Fossil budgets, company structures, national election results

Stacked column/bar
A simple way of showing part-to-whole relationships but can be difficult to read with more than a few components.

Marimble
A good way of showing the size and proportion of data at the same time – as long as the data are not too complicated.

Pie
A common way of showing part-to-whole data – but be aware that it's difficult to accurately compare the size of the segments.

Donut
Similar to a pie chart – but the centre can be a good way of making space to include more information about the data (eg labels).

Tree map
Use for hierarchical part-to-whole relationships. Can be difficult to read when there are many small segments.

Venn
A Venn diagram, often used for illustrating comparative composition by number of sets.

Gridmap
Good for showing 5 information that work best when used on whole numbers and with small, multiple layout forms.

Venn
Generally only used for schematic representation.

Waterfall
Can be useful for showing part-to-whole relationships where some of the components are negative.

Spatial

Aide from locator maps only used where precise location or geographical patterns in data are more important to the reader than anything else.

Example FT uses
Natural resource locations, natural disaster risk/impact, government areas, election in election results

Basic choropleth (contour)
The standard approach for putting data on a map – should always be used where the data are more important than the map itself.

Proportional symbol (contour/magnitude)
Use for totals, rather than rates – as they are more difficult to see.

Flow map
For showing movement across a map.

Contour map
For showing areas of equal value on a map. Can use different colour schemes for showing ranges with equal value.

Equalised cartogram
Correcting each unit on a map to a regular and equally-sized shape – good for representing varying regions with equal value.

Scaled cartogram (inset)
Stretching and shrinking a map so that each area is sized according to a particular value.

Dot density
Used to show the location of individual points. Make sure to include a scale to indicate the number of items the reader should see.

Heat map
Grid-based data values mapped with an arbitrary colour scale. A choropleth map is not mapped to an administrative unit.

Flow

Show the reader volumes or intensity of movement between two or more states or conditions. These might be signal sequences or geographical locations.

Example FT uses
Movement of funds, trade, migration, leavens, information, relationship graphs

Sankey
Shows changes in flows from one condition to at least one other – good for tracing the overall outcome of a complex process.

Waterfall
Designed to show the sequencing of data through a flow process. Usually budgets. Can include +/- components.

Chart
A complex but powerful diagram which can show a flow through a process. Can include +/- components.

Network
Used for showing the strength and relationships of varying types.

Visual vocabulary

Designing with data

There are so many ways to visualise data - how do we know which one to pick? Use the categories across the top to decide which data relationship is most important in your story, then look at the different types of chart within the category to form some initial ideas about what might work best. This list is not meant to be exhaustive, nor a wizard, but is a useful starting point for making informative and meaningful data visualisations.

FT graphics: Alex Smith, Chris Campbell, Ian Bell, Liz Evans, Stephen Francis, Alex Henderson, Thomas Ford, Nick O'Brien, Martin O'Brien, inspired by the Graphics Community by Jon Schroeder and Sara K. Berman

[ft.com/vocabulary](https://www.ft.com/vocabulary)



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Final Thoughts

- Visualisation Plan, **slide 3**: *how will you make sure your dashboard doesn't mislead?*
- Answer it using these rules, with your own data and your own charts
- Name the rule, show the risk, say what you'll do about it

Next week: Checkpoint 1

- Thu 15 Oct (Topic 4 lab), with a second slot on Thu 22 Oct (Topic 5 lab)
- Submit (**deadline: 15th October, 1 pm**): one or two sentences on your personal dataset, the columns, and some rows already collected (or where the export will come from)
- Counting something yourself? You need six weeks. **(Ideally) Start this week!**

Lab
